Research Article

SBA Loan Default Prediction Using Gradient-Boosted Trees and a Profit-Maximizing Approval Rule

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Abstract

We study default prediction for U.S. Small Business Administration (SBA) 7(a) loans using 866,082 observations enriched with seven external datasets yielding 15 economic indicators merged at loan origination. We compare 21 model configurations across logistic regression, discriminant analysis, random forest, bagging, AdaBoost, gradient boosting, and multi-layer perceptrons under an asymmetric lending payoff (+5% gain, −25% loss per the CSUS 2024 Business Analytics Competition brief). A 30-trial Optuna search on the winning LightGBM family produces the recommended model: an Optuna-tuned configuration (418 trees, 194 leaves) achieving $1,096.4M in validation profit at 77.8% approval with AUC 0.9827 and 1.09% approved default rate. The best pre-specified benchmark entry (LightGBM n300 nl127) reaches $1,078.6M. Relative to an approve-all baseline of $483.9M, the recommended rule captures $612.5M by screening out the riskiest applicants. A staged ablation shows that external economic indicators add interpretable risk channels (housing stress, cost pressure, disaster exposure) and an $8.5M profit gain over the SBA-only baseline.

Keywords: *SBA loans, credit risk, default prediction, LightGBM, gradient boosting, profit-based threshold, Optuna, gains and lift charts*

1. INTRODUCTION

The U.S. Small Business Administration 7(a) program extends credit to firms that may not obtain conventional financing on comparable terms. For lenders, the program creates a recurring screening problem: approving a sound borrower generates a modest return, whereas approving a borrower who later defaults can erase the gains from many successful loans. In that setting, a useful credit model is not simply one that ranks risk well, but one that supports a lending rule aligned with the economics of approval and denial.

This problem is especially difficult for small-business lending. Compared with large public firms, small firms disclose less information, differ more widely in business structure, and are more exposed to local shocks. A model based only on a narrow set of linear relationships may therefore miss interactions between borrower characteristics, loan design, industry conditions, geography, and the economic environment at origination.

Our study builds the problem around the lender's actual decision point. We start from the SBA application file, merge seven external datasets using only information available at or before approval, compare a broad set of model families under a profit-based objective, and translate the selected model into an approval rule through threshold tuning, profit curves, and gains-and-lift diagnostics.

The paper makes two linked contributions. Empirically, it tests whether approval-time external context improves a large public SBA credit-risk workflow. Operationally, it expresses the result in the form a lender would use: a model choice, a probability-of-default cutoff, and the resulting approval rule.

The remainder of the paper proceeds as follows. Section 2 reviews the literature on SME default prediction, external credit-risk information, and machine-learning evaluation. Section 3 describes the data, merge rules, and profit-based methodology. Section 4 presents the exploratory evidence that motivated feature design. Section 5 summarizes the model families considered. Section 6 reports the validation and test results and states the final approval rule. Section 7 discusses interpretation, limitations, and implications.

2. LITERATURE REVIEW

2.1. SME and SBA Loan Default Prediction

Credit-risk modeling for small and medium-sized enterprises differs from consumer credit scoring and large-firm bankruptcy prediction. Large listed firms typically provide audited financial statements, market prices, analyst coverage, and long operating histories. Small firms are far more opaque. They often have limited disclosure, short performance records, concentrated revenue sources, and substantial exposure to local shocks. In that setting, default prediction is less about identifying one dominant accounting ratio than about combining multiple imperfect indicators of fragility.

Ciampi et al. [3] make this point clearly in their systematic review of SME default prediction. They show that the field has moved steadily beyond traditional accounting ratios toward broader predictor sets that include qualitative variables, macroeconomic conditions, governance, and other contextual signals. Malakauskas and Lakstutiene [4] and Altman et al. [5] reach similar conclusions from empirical work: predictive performance improves when models incorporate firm age, sector, scale, organizational characteristics, and other information that goes beyond a narrow balance-sheet view. This literature is directly relevant to the SBA setting, where many borrowers do not come with rich financial statements and the model must instead learn from application-level information such as loan size, term, employee count, business age, industry, geography, and guarantee structure.

The closest predecessor to our study is Li et al. [1], who introduced the SBA 7(a) dataset as an approve-versus-deny decision problem. That paper established two points that remain central here: the SBA records already contain strong economic risk indicators, and the task should be framed as a lending decision rather than as a generic classroom classification exercise. At the same time, Li et al. [1] remain pedagogical in scope: the analysis centers on logistic regression, uses misclassification and a simple cutoff rule to evaluate decisions, and explicitly notes concerns about time-window selection bias and the

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absence of denied applications from the observed sample. Our study builds on that foundation but moves it into a research setting by comparing broader model families, introducing approval-time external context, and aligning the final approval rule with profit rather than equal-cost classification error.

2.2. External Economic Context at Origination

A second stream of literature asks what is missing from the borrower record itself. Carvalho et al. [2] show that macroeconomic conditions materially influence default risk even after controlling for firm-specific characteristics. Gallas et al. [14] reach a related conclusion from a stress-testing perspective, showing that inflation, unemployment, and broader macro shocks can alter portfolio risk even when the underlying borrower pool is unchanged. In this literature, credit quality depends on the approval environment as well as on the borrower profile.

Other work extends the information set through nontraditional or higher-frequency data. Crosato et al. [10] show that website-based information can improve credit-risk modeling when accounting variables are incomplete or slow-moving. Peng et al. [16] find that transaction behavior and market-evaluation data contain early signals of SME distress that conventional records do not fully capture. Although our project does not use website or transaction data, the underlying idea is the same: small-business risk is partly revealed through operating context rather than through borrower characteristics alone.

This literature supports the inclusion of external variables, but it also creates an important design constraint. In an origination setting, the lender can only use information observable at or before approval. Many studies test macroeconomic effects in firm panels or bank portfolios without having to reproduce that approval-time restriction. Our study is built around that constraint. We merge housing-market conditions, inflation, oil shocks, personal income changes, bank failures, disasters, and labor-market indicators only through lagged or look-back windows that preserve the lender's information set at the time of decision.

2.3. Machine Learning for Nonlinear Credit-Risk Structure

The modeling literature has likewise moved beyond the assumption that logistic regression should be the default endpoint. Rogojan et al. [6] document the shift from traditional linear scoring models toward ensemble methods and neural approaches. Chang et al. [9], Aruleba and Sun [7], Emmanuel et al. [11], and Sun et al. [17] show that boosted ensembles, stacked classifiers, and other flexible methods can improve credit-risk prediction when relationships are nonlinear and interactions are important.

That point is especially relevant for SBA loans. The effect of one variable may depend heavily on another: a construction borrower may become far riskier when house prices are falling; a micro business may be more vulnerable when inflation is high; a tourism-oriented firm may respond differently to disaster exposure than a business in a less shock-sensitive sector. Tree-based models are particularly attractive in this setting because they can represent thresholds, segment-specific relationships, and high-order interactions without imposing one global linear effect for each variable.

Bosker et al. [12] add another relevant insight by emphasizing clustered credit-risk structure. Default can be correlated within industries, regions, and borrower groups rather than being independently generated observation by observation. That clustering is central to SBA data, where state, NAICS sector, lender geography, and approval era all plausibly induce dependence in default patterns. This provides a strong methodological rationale for comparing linear, tree-based, and neural models rather than assuming that a single additive specification can capture the structure of the problem.

2.4. Profit-Based Evaluation and Threshold Choice

Even when the prediction model is well chosen, an additional question remains: how should the lender use the predicted probabilities? Much of the credit-risk literature still emphasizes AUC, accuracy, recall, or F1-score. These are useful descriptive metrics, but they do not map directly to the economics of approval decisions. In lending, approving a defaulted loan and rejecting a repaid loan are not symmetric mistakes.

Cui et al. [13] argue explicitly for profit-oriented loan default prediction, showing that lender return should guide model evaluation rather than symmetric error metrics alone. Kpatcha [15] likewise shows that threshold selection is not a minor technical detail: the approval cutoff changes both economic outcomes and the profile of borrowers who receive credit. Relative to this literature, our project takes the decision rule seriously. We do not stop at estimating default probabilities and reporting AUC. We convert model output into an approval policy by tuning each model's cutoff on the validation set using the lender's payoff structure.

This distinction is also what separates our work most sharply from the original SBA classroom paper. Li et al. [1] treat the cutoff largely as a classification choice and evaluate performance through misclassification. Our objective is different. We define an asymmetric payoff matrix for approval decisions, derive the implied break-even default probability, and then allow the empirical threshold to be chosen by validation profit rather than by equal-cost error minimization.

2.5. Positioning of This Study

This literature leaves a clear opening for the present study. The SBA setting calls for a reproducible public dataset, approval-time external information, and a decision criterion that follows lender profit rather than symmetric classification error. We bring those elements together in one workflow: a large SBA origination file, external context merged under a strict timing rule, a broad family comparison, and final threshold selection based on profit.

3. DATA AND METHODOLOGY

3.1. The SBA Loan Dataset

The dataset comes from the SBA's public data portal and contains 866,082 loans originated between 1987 and 2014 across 100 columns. The broader SBA loan records were previously introduced to the statistics-education literature by Li et al. [1]; our study extends that foundation from a classroom logistic-regression case to a full-scale credit-risk workflow. Our target $y \in \{0, 1\}$ codes whether the loan paid in full ($y = 0$) or was charged off ($y = 1$). The overall default rate is 17.27%.

One variable merits special mention. `DisbursementGross` records the amount actually disbursed after loan approval. That figure is unavailable at origination, so including it would leak post-decision information into the model. We exclude it from all predictors.

We assign each loan an era based on its approval year: `pre` (1987–2007, $n = 782{,}360$, 14.3% DR), `crisis` (2008–2009, $n = 50{,}401$, 29.3% DR), and `post` (2010–2014, $n = 33{,}321$, 8.7% DR). The post-crisis era is small (3.8% of loans) and right-censored, so post-era results must be interpreted cautiously.

3.2. Leakage Audit

Before model fitting, we audit the candidate predictors for target leakage and post-origination information. Table 2 lists the variables excluded from model inputs and the reason for each exclusion. The target variable `MIS_Status` is never used as a predictor. `ChgOffDate`, `ChgOffPrinGr`, and `BalanceGross` are post-default outcome fields and are removed. `DisbursementGross` records the realized amount disbursed after approval, so it is excluded from predictors and retained

Table 1. Key SBA variables used in modeling. Variables shown in their final form after preprocessing (see Section 3.5).

Category	Variables	Description
Loan Terms	Term, log_GrAppv, Portion	Duration in months, log gross amount, SBA guarantee ratio
Borrower	CreateJob, RetainedJob, log_NoEmp, NewExist, UrbanRural	Jobs created/retained, log employees, business age, urban/rural location
Industry	NAICS_sector	2-digit NAICS industry code
Lender	State, BankState, same_state_bank	Borrower state, bank state, local lending indicator
Program	LowDoc, RevLineCr, RealEstate, jobs_per_dollar	Doc type, credit line, real estate backing (Term≥240), employment efficiency

only as the retrospective exposure amount used in profit calculations on validation and test sets. The lender identifier `Bank` is a high-cardinality categorical field (>5000 unique values); we keep the geographic information through `BankState` and `same_state_bank` without encoding lender identity directly.

Table 2. Leakage audit: variables excluded from model features.

Variable	Status	Reason
MIS_Status	Excluded	Target variable
ChgOffDate	Excluded	Post-default date
ChgOffPrinGr	Excluded	Post-default charge-off amount
BalanceGross	Excluded	Post-performance balance
DisbursementGross	Excluded	Realized amount unknown at origination; retained only as the profit exposure weight
Bank	Excluded	High-cardinality lender identity; replaced by BankState and same_state_bank

Seven external datasets were merged with the SBA data. The critical constraint is temporal: every external variable reflects conditions *before* the loan was approved, using the loan's approval date as a reference point for lookback windows.

Table 3. External data sources with access details and join rules.

Source	Variables	Key	Window
FHFA HPI	hpi_flag, growth, drawdown	State×Year	12 mo
BLS CPI	high_infl_flag, cpi_yoy	Natl×Year	12 mo
EIA Petroleum	oil_shock_flag, oil_yoy	Natl×Year	12 mo
BEA Income	income_decl_flag, growth_1y	State×Year	Prior yr
FDIC Failures	bank_failures_12m, bank_failures_24m	State×Date	12–24 mo
FEMA Disasters	hurricane, fire, flood, storm flags	State×Date	12 mo
BLS QCEW	emplvl_chg, wages_chg, avg_wage	State×Year	Prior yr

Public source portals: fhfa.gov, bls.gov/cpi, eia.gov, bea.gov, fdic.gov, fema.gov/open, and bls.gov/cew.

The FEMA join is the most granular. For each loan, we count disaster declarations in the borrower's state during the 365 days preceding the approval date, producing separate binary flags for hurricanes, floods, fires, and severe storms. The `high_inflation_flag` and `oil_shock_flag` are threshold-based: each is 1 if the year-over-year change in CPI (or crude petroleum PPI) in the 12 months before origination exceeds its 90th percentile across the full sample period. This transforms continuous economic data into binary distress signals; we justify the binary encoding in Section 4.

3.3. Split Strategy

All splits use `random_state = 1`. We employ a two-step stratified split. First, an 80/20 outer split divides the full dataset of 866,082 loans into a train-validation set of 692,865 loans and a held-out test set of 173,217 loans. Second, an 80/20 inner split on the train-validation set produces a training set of 554,292 loans and a validation set of 138,573 loans. Both splits are stratified by $y \times$ era, ensuring that pre-crisis, crisis, and post-crisis originations are proportionally represented in every partition. The validation set selects the model and tunes the threshold; the held-out test set is reserved exclusively for final confirmation of the selected model and threshold.

3.4. The Profit Function

Let a_i denote the loan amount for application i, and let $d_i \in \{0, 1\}$ be the lender's decision (1 = approve, 0 = deny). Given the true outcome y_i, the profit from that single decision is:

$$\Pi(d_i, y_i, a_i) = \begin{cases} +0.05\, a_i & \text{approve, borrower repays} \\ -0.25\, a_i & \text{approve, borrower defaults} \\ 0 & \text{deny} \end{cases} \tag{1}$$

These percentages follow the CSUS 2024 Business Analytics Competition brief [8]. The 5% gain approximates the net interest margin on a typical SBA-guaranteed loan after servicing costs. The 25% loss reflects the severity of an unsecured small business default, where recovery rates are low and collection costs are high. The 5:1 loss-to-gain ratio means that a single default wipes out the profit from five repaid loans of equal size. This asymmetry shapes every decision rule we derive.

The break-even default probability follows directly: set expected profit $0.05(1 - p^*) - 0.25p^* = 0$, yielding $p^* = 1/6 \approx 0.1667$. If a model predicts a 17% or higher chance of default, approving the loan loses money in expectation. But this theoretical threshold assumes perfect calibration, which no model achieves. We therefore select the threshold empirically, scanning $\tau \in [0.01, 0.60]$ at 120 points and choosing the value that maximizes $\sum_i \Pi(\mathbf{1}[\hat{p}_i \le \tau], y_i, a_i)$ on the validation set.

We report area under the ROC curve (AUC) for ranking quality, Brier score for probability calibration, net profit as the primary optimization target, approval rate, and approved default rate. Accuracy, precision, recall, and F1-score are computed but not optimized, because they treat false approvals and false denials identically. That symmetry is at odds with the asymmetric cost structure of lending.

3.5. Preprocessing

The raw SBA application variables require several transformations before entering a model pipeline. Each step is motivated by the statistical properties of the underlying data or by solver requirements.

Log transformations. Loan amounts and firm size are right-skewed: a small number of multi-million-dollar loans and large employers sit far from the median. For linear models and MLPs, this skew distorts gradient-based optimization because a handful of extreme values dominate the loss. We apply $\log(1 + x)$, which compresses the right tail while preserving zero-valued observations:

$$\texttt{log\_GrAppv} = \ln(1 + \texttt{GrAppv}),$$
$$\texttt{log\_SBA\_Appv} = \ln(1 + \texttt{SBA\_Appv}),$$
$$\texttt{log\_NoEmp} = \ln(1 + \texttt{NoEmp}).$$

Tree-based models are invariant to monotonic transformations, so the log versions are neutral for those families but essential for fair comparison across model classes on standardized inputs.

Derived financial ratios. Five structural features are computed from SBA application fields to capture loan structure and employment efficiency:

$$\texttt{Portion} = \texttt{SBA\_Appv}/\texttt{GrAppv},$$
$$\texttt{unguaranteed\_ratio} = 1 - \texttt{Portion},$$
$$\texttt{unguaranteed\_amount} = \texttt{unguaranteed\_ratio} \times \texttt{GrAppv},$$
$$\texttt{jobs\_total} = \texttt{CreateJob} + \texttt{RetainedJob},$$
$$\texttt{jobs\_per\_dollar} = \texttt{jobs\_total}/\texttt{GrAppv}.$$

`Portion` is the SBA guarantee share; a higher share shifts default risk toward the SBA and may influence bank underwriting standards. `jobs_per_dollar` captures employment-generation efficiency per dollar lent, a metric that regulators and policymakers track for the 7(a) program.

Two of these ratios are linearly dependent on their source variables: `unguaranteed_ratio` $= 1 -$ `Portion` and `unguaranteed_amount` $= (1 -$ `Portion`$) \times$ `GrAppv`. Both are fully determined by `Portion` and `GrAppv` and are removed during multicollinearity screening, along with `jobs_total` which is the sum of `CreateJob` and `RetainedJob`. We retain `Portion` because the SBA guarantee share is the more commonly reported metric, and we retain both `CreateJob` and `RetainedJob` because a business creating new positions differs from one preserving existing jobs. `jobs_per_dollar` is kept because it captures employment efficiency per dollar lent.

Binary indicator variables. `RealEstate` $= 1$ when `Term` ≥ 240 months, identifying loans likely backed by real estate collateral with longer amortization schedules. `same_state_bank` $= 1$ when the borrower's state matches the originating bank's state, capturing local lending relationships where the bank has better information about the borrower and local economy.

Era stratification variable. We assign each loan to a macroeconomic era based on its approval year: `pre` (1987–2007, 90.3% of loans, 14.3% DR), `crisis` (2008–2009, 5.8% of loans, 29.3% DR), and `post` (2010–2014, 3.8% of loans, 8.7% DR). This variable is used only for stratified data splitting and as a diagnostic grouping variable – it is not included as a model feature.

Excluded temporal features. We exclude the raw calendar-year variables `approval_year` and `ApprovalFY_clean` from all models. The multicollinearity screening described below shows these variables are collinear with the external macro indicators (HPI growth, CPI, oil prices, personal income) that carry the same temporal signal through economic channels. A raw year variable memorizes which calendar periods had high default rates rather than learning transferable risk patterns.

Engineered interaction features. The exploratory analysis in Section 4 identifies thirteen conditional risk signals where the effect of an external condition depends on a borrower or geographic characteristic. We construct these as binary products. Two representative examples:

$$\texttt{construction\_hpi} = \mathbb{1}[\texttt{NAICS} = 23] \times \mathbb{1}[\texttt{hpi\_neg} = 1],$$

$$\texttt{tourism\_hurricane} = \mathbb{1}_{\mathcal{T}} \times \texttt{fema\_hurricane\_12m},$$

where $\mathbb{1}_{\mathcal{T}}$ indicates tourism-sector NAICS codes $\{44, 45, 71, 72\}$. The first isolates construction firms during housing downturns; the second captures tourism-sector borrowers after recent hurricane declarations. The complete set of 28 engineered and derived variables, with formulas and rationales, is provided as supplementary material.

Encoding and final dimensionality. After multicollinearity screening, the final modeling matrix contains 37 numeric and 7 categorical variables (44 total). For tree-based models, the categorical variables are ordinal-encoded, so the learner reads the 44-column matrix directly. For linear and MLP models, the same 7 categoricals are one-hot encoded, expanding the design matrix to 183 columns. All numeric features receive median imputation. Linear and neural models additionally receive z-score standardization, which is required for ℓ_2-regularized logistic regression and accelerates SGD convergence in MLPs by providing features on a common scale.

Multicollinearity screening and feature reduction. Before model training, we conducted a systematic multicollinearity analysis on the full set of 54 candidate features (45 SBA-derived plus 9 external). The Pearson correlation matrix revealed several near-perfect associations: raw and log-transformed loan amounts correlated above $r = 0.97$, employment counts and their log transforms exceeded $r = 0.95$, and the SBA guarantee ratio (`Portion`) and unguaranteed ratio summed identically to 1, producing $r = -1.00$. These redundancies inflate variance for linear models and add no signal for any model family.

We computed the Variance Inflation Factor (VIF) for all numeric features. Using a threshold of VIF > 10, ten features were removed. Raw loan amounts (`GrAppv`, `SBA_Appv`) and their log transforms were near-perfectly collinear ($r > 0.97$); we retained only `log_GrAppv`. `NoEmp` was dropped in favor of `log_NoEmp` for the same reason. `unguaranteed_ratio` is the exact complement of `Portion` ($r = -1.00$) and was removed. `unguaranteed_amount` is fully determined by `Portion` and `GrAppv`. `jobs_total` is the sum of `CreateJob` and `RetainedJob`; we dropped the aggregate and kept the components because a business creating 10 new jobs differs economically from one retaining 10 existing positions. The raw calendar-year variables `approval_year` and `ApprovalFY_clean` were removed because they memorize cohort labels rather than learning transferable risk patterns. `state_bank_failures_24m` correlates at $r = 0.93$ with its 12-month counterpart and was dropped.

After removing these ten variables, the maximum pairwise correlation among remaining features falls below 0.85. Several retained features (`Portion`, `log_GrAppv`, `CreateJob`, `RetainedJob`) carry elevated VIF individually but encode distinct economic information that tree-based models can use without inverting a covariance matrix. The benchmark leaderboard in Section 6.1 is built on this cleaned 44-feature set.

Full 37-feature correlation. Fig. 1 shows the complete Pearson correlation matrix across all 37 numeric features. External macro indicators and borrower-level SBA features are essentially orthogonal: the median absolute correlation across the entire 37-feature block is 0.043, and no SBA-to-external pair exceeds $|r| = 0.15$. Table 4 lists the top 15 strongest pairwise correlations for reference. The strongest correlation is between `Term` and `RealEstate` ($r = +0.869$), reflecting the structural definition `RealEstate` $= \mathbb{1}[$`Term` $\geq 240]$. Engineered interactions show moderate correlations with their component variables ($|r| \approx 0.54$–0.85), as expected from binary-product construction.

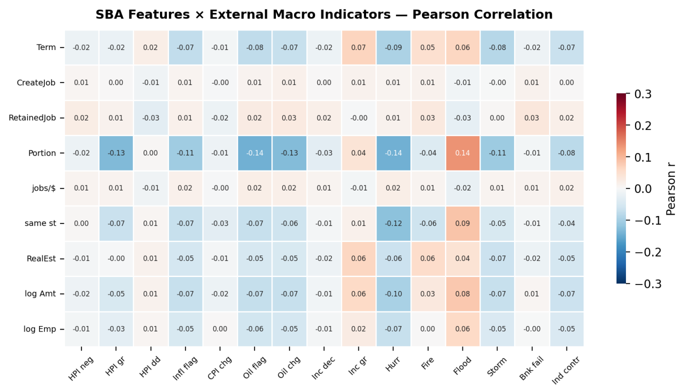

Figure 1. Pearson correlation between 9 SBA-derived features (rows) and 15 external macro indicators (columns). No pair exceeds $|r| = 0.15$, confirming that external data captures economic context independent of borrower-level application characteristics. Values shown in each cell.

4. EXPLORATORY DATA ANALYSIS AND FEATURE DESIGN

4.1. Borrower Fragility and Macro Stress

The original SBA application variables reveal a consistent pattern: shorter loan terms, smaller loan amounts, and fewer employees are all associated with higher default rates. Fig. 2 shows the gradient across each dimension. Loans under 60 months default at 38.2%, more than ten times the rate of loans over 180 months (3.0%). Loans under \$50,000 default at 26.5%, while loans of \$200,000 or more default at 8.7%. Firms with 0–1 employee default at 22.8%, compared with 7.8% for firms with 21 or more employees. Each pattern is monotonic and economically interpretable: short-term loans are typically working-capital facilities with limited collateral, small loans reach the most

Table 4. Top 15 strongest pairwise Pearson correlations across the full 37-numeric-feature set, sorted by $|r|$ descending.

Rank	Feature A	Feature B	r	$\|r\|$
1	`Term`	`RealEstate`	+0.8688	0.8688
2	`Hurricane 12m`	`Coastal×Hurr`	+0.8476	0.8476
3	`High-infl flag`	`Infl×Micro`	+0.7615	0.7615
4	`Oil-shock flag`	`Oil YoY chg`	+0.7511	0.7511
5	`Tourism×Hurr`	`Disaster×Hurr`	+0.7236	0.7236
6	`Hurricane 12m`	`Disaster×Hurr`	+0.6748	0.6748
7	`HPI↓ flag`	`HPI drawdown 2y`	−0.6659	0.6659
8	`High-infl flag`	`CPI YoY chg`	+0.6643	0.6643
9	`Tourism flag`	`Disaster-sens flag`	+0.6413	0.6413
10	`Income↓ flag`	`Income growth 1y`	−0.6138	0.6138
11	`HPI↓ flag`	`HPI growth 1y`	−0.6096	0.6096
12	`HPI growth 1y`	`HPI drawdown 2y`	+0.5931	0.5931
13	`Term`	`log(GrAppv)`	+0.5750	0.5750
14	`Coastal×Hurr`	`Disaster×Hurr`	+0.5412	0.5412
15	`HPI drawdown 2y`	`Bank-fail 12m`	−0.5001	0.5001

Computed from 866,082 enriched SBA loans. Among 666 pairwise correlations across 37 numeric features, only 15 exceed $|r| > 0.5$ *and only 5 exceed* $|r| > 0.7$*. The median absolute correlation is* 0.043*. Shortened feature names (see Appendix C for full definitions).*

fragile borrowers, and micro-businesses operate with thin cash buffers and concentrated revenue streams.

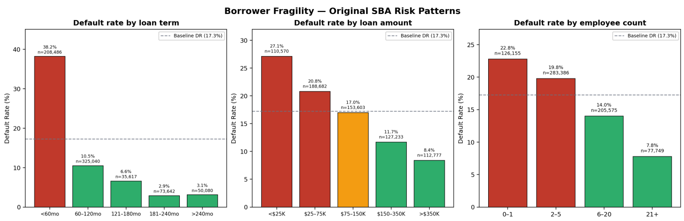

Figure 2. Borrower fragility: default rates by term, loan size, and number of employees. All three patterns are monotonic. The dashed reference line marks the overall training-set default rate of 17.3%.

These patterns raise a natural question: if small, short-term borrowers are already riskier under normal conditions, do external cost shocks amplify that fragility? We test two macro-stress channels: inflation and oil prices.

Fig. 3 (left panel) shows the interaction of inflation with firm size. Under normal inflation conditions, default rates are graded across firm sizes, with micro firms (0–5 employees) at 18.1% and large firms (21+ employees) at 7.1%. Under high inflation (defined as a year-over-year CPI change exceeding the 90th percentile of the historical distribution) micro-firm defaults jump to 30.8%, a 12.7 percentage-point increase, while large-firm defaults rise to only 11.7%, a much smaller 4.6-point increase. This is consistent with micro-businesses lacking pricing power and cost-pass-through ability: when input costs rise, their margins compress faster than they can adjust prices.

The oil shock interaction is reported in the supplementary tables. Firms in fuel-sensitive sectors (transportation, construction, manufacturing) show elevated default rates under both normal and oil-shock conditions compared with non-exposed firms, but the *relative* increase under an oil shock is broadly similar across exposed and non-exposed sectors, suggesting that oil price spikes capture a general macro-stress signal rather than a narrow transportation-sector channel. The macro-stress signals are therefore interpreted as broad cost-pressure indicators, not as narrowly targeted sector effects.

Bank failures and income decline, in contrast, show patterns that are more consistent with selection and lending-tightening effects than with direct causal stress. Loans originated in states with recent bank failures (within the prior 12 months) have a *lower* observed default rate (17.0%) than loans in states without recent failures (26.4%). This is unlikely to mean bank failures reduce risk. It more likely reflects that lenders and the SBA tightened underwriting standards in affected states, so the loans that were approved in those conditions were drawn from a more carefully screened pool. The income-decline flag shows a similar pattern: consumer-facing sectors have lower observed defaults under income-decline conditions than under stable-income conditions, which again points to a selection mechanism (only the strongest consumer-facing borrowers receive credit when local purchasing power is falling) rather than a causal protective effect.

These findings motivate four engineered features. `inflation_micro_business` captures the disproportionate vulnerability of the smallest firms to cost pressure. `oil_shock_supply_chain` flags fuel-exposed sectors during oil-price spikes. `income_decline_consumer_sector` and `recent_bank_failure_flag` are retained as credit-cycle and regime indicators, but we interpret them cautiously: they signal when underwriting standards are likely to be tighter, not that income decline or bank failures directly cause higher default.

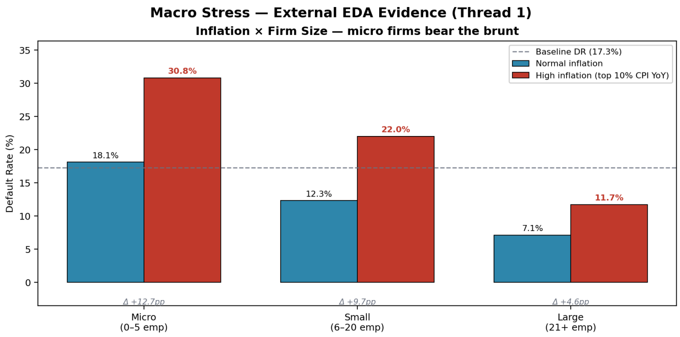

Figure 3. Macro stress and housing: inflation interacts strongly with firm size (left), while falling HPI amplifies construction-sector risk (right). The dashed reference line marks the 17.3% overall training-set default rate.

4.2. Industry Cyclicality and Local Business Environment

Default risk is not evenly distributed across industries. Fig. 4 (left panel) ranks NAICS sectors by default rate. Real estate and rental/leasing (NAICS 53, 28.9%), finance and insurance (NAICS 52, 28.4%), transportation and warehousing (NAICS 48, 27.1%), and information (NAICS 51, 25.2%) lead the default table, while healthcare and social assistance (NAICS 62, 10.4%) and the unknown/missing NAICS group (5.6%) sit at the bottom. The sectors near the top are those most exposed to asset-price cycles, fuel costs, and discretionary consumer spending.

Two additional baseline signals deserve mention. New businesses (NewExist = 2) default at 18.3%, only modestly higher than existing businesses at 16.9%, a gap of 1.4 percentage points. While the direction is consistent with the pattern that lenders face more information asymmetry when underwriting firms without an established operating history, the magnitude is small, and this variable should be interpreted as a modest supporting signal rather than a major risk driver. A much stronger signal comes from lender geography: same-state lending (lender and borrower in the same state) has a default rate of 9.9%, while out-of-state lending reaches 25.3%. This 15.4 percentage-point gap indicates that local lenders possess soft information that allows them to underwrite more effectively, while out-of-state lenders may lend only to higher-risk applicants who cannot obtain financing locally, consistent with information asymmetry in out-of-state lending.

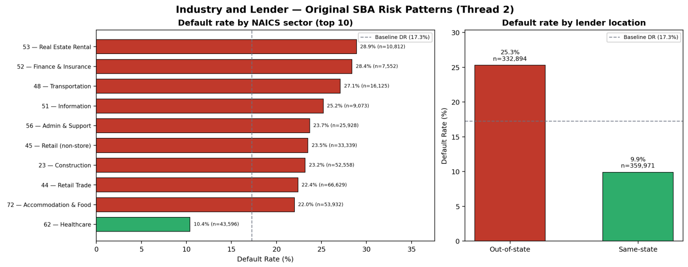

Figure 4. Industry and lender risk patterns. Left: top NAICS sectors by default rate, with healthcare shown as a low-risk reference. Right: same-state lending is associated with substantially lower default rates than out-of-state lending.

The industry risk patterns prompt a hypothesis: high-default sectors are not universally risky; their risk intensifies when local economic conditions deteriorate. Fig. 3 (right panel) tests this for the construction sector. Construction firms in states where the house price index was *not* declining in the year before origination default at 22.5%. When HPI is falling, the construction default rate rises to 26.5%, while non-construction firms in falling-HPI states default at 21.0%. The important result is that both the sector effect and the HPI effect are present, and the interaction (construction + falling HPI) produces the highest observed default rate. The same qualitative pattern holds for real estate firms (NAICS 53), where falling HPI elevates default rates from 27.2% to 35.9%, as shown in the supplementary tables.

Fig. 5 extends this logic to supply-chain-sensitive sectors under high inflation. Non-supply-chain firms in normal inflation conditions default at 14.3%, while supply-chain-sensitive firms under high inflation default at 28.6%. The interaction is strong but, as with the oil-shock analysis in the previous subsection, the main effect of inflation is broad: the absolute increase for non-supply-chain firms (from 14.3% to 25.9%, +11.6 points) is comparable to the increase for supply-chain firms (from 17.9% to 28.6%, +10.7 points). This pattern reinforces the interpretation of inflation as a general cost-pressure signal rather than a narrow sector-specific vulnerability.

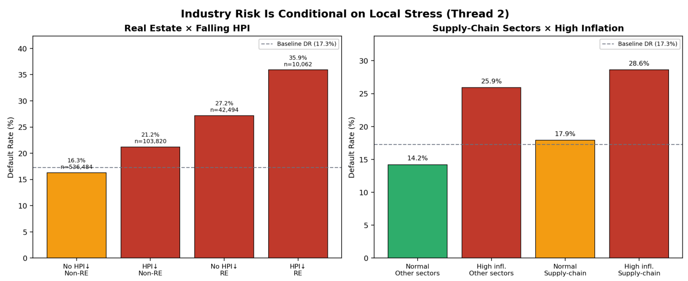

Figure 5. Local market stress: real estate sectors are more vulnerable to falling HPI (left), while inflation elevates default rates broadly, with supply-chain sectors starting from a higher baseline (right).

The QCEW state-industry contraction data provides additional local-ecosystem context. When a state-industry cell registers negative employment growth in the year before origination, default rates in that cell are elevated relative to cells with stable or growing employment. The `industry_contraction_flag` is retained in the final feature set; the remaining QCEW variables (employment level change, wage growth, average weekly wage) were excluded because their incremental predictive contribution was modest and their temporal granularity is coarser than the daily origination events they aim to contextualize.

From this analysis, three features are constructed: `construction_hpi` and `real_estate_industry_hpi` capture the interaction of construction and real estate sectors with falling house prices, and `supply_chain_inflation` flags supply-chain-exposed sectors under high inflation. The QCEW state-industry contraction data is retained as exploratory diagnostics; its temporal granularity is coarser than the daily origination events it contextualizes.

4.3. Geography, Disaster Exposure, and Tourism Vulnerability

Default risk is geographically concentrated. Fig. 6 (left panel) shows the fifteen states with the highest default rates. Florida (27.3%), Georgia (23.9%), Nevada (23.4%), and Michigan (22.6%) occupy the top positions. These states share a combination of housing-intensive economies, tourism dependence, and exposure to natural disasters. The right panel confirms that tourism-related sectors (NAICS 44, 45, 71, 72), coastal states, and disaster-sensitive sectors (construction, retail, transportation, information, accommodation) each carry elevated default risk relative to their non-exposed counterparts.

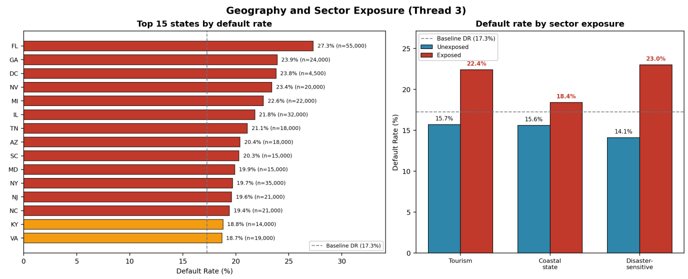

Figure 6. Geography and sector exposure: the 15 highest-default states (left) and default-rate differences by tourism, coastal, and disaster-sensitive sector classification (right).

The geographic clustering of default risk raises the question of whether physical hazards amplify it. The external FEMA data allow us to test whether natural disasters matter and, if so, whether all disasters are alike.

Fig. 7 provides a clear answer: disaster type matters sharply. Hurricane exposure produces the largest positive default-rate gap (+11.3 percentage points, from 15.5% to 26.9%). Fire, severe storms, and emergency declarations show positive but smaller gaps (+6.5, +5.4, and +2.8 points, respectively). At the opposite end, flood, drought, tornado, and earthquake exposure are associated with *lower* observed default rates than non-exposed loans (flood: −10.8 points; drought: −14.2 points). These reversed patterns are not evidence that floods or earthquakes protect borrowers. They are more plausibly explained by selection effects: flood-zone construction requires insurance and stricter building codes; earthquake exposure is concentrated in a few states with well-known seismic risk and hardened infrastructure; drought and tornado declarations cover very small geographic footprints and too few loans to produce reliable statistical averages. The negative deltas should be interpreted as evidence of differential preparedness and underwriting adjustments, not as causal risk-reducing effects of the disasters themselves.

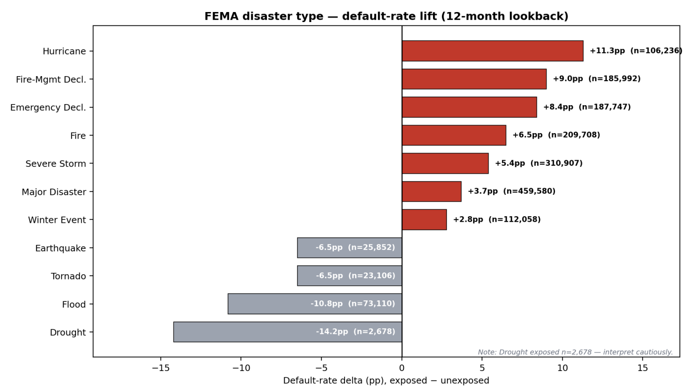

Figure 7. FEMA disaster type default-rate deltas (exposed minus unexposed). Positive deltas in red indicate higher default rates among exposed borrowers; negative deltas in gray are likely driven by selection and preparedness effects.

Because hurricanes show the largest and most robust positive association, we probe the interaction of hurricane exposure with sector and geographic categories. Fig. 8 displays three interactions. Tourism businesses exposed to a hurricane in the prior twelve months default at 31.2%, compared with 20.6% for unexposed tourism firms and 14.0% for unexposed non-tourism firms. The hurricane effect is present for both tourism and non-tourism borrowers, but tourism firms carry a higher baseline risk even without hurricane exposure.

The coastal-hurricane interaction shows a more nuanced pattern. Non-coastal borrowers exposed to a hurricane default at 29.0%, the highest single-group rate in this panel. This is consistent with the pattern that inland hurricane declarations capture events where a hurricane's path causes inland flooding and wind damage that catches unprepared borrowers, whereas coastal communities have adapted building codes and evacuation protocols that mitigate, though do not eliminate, hurricane-driven business disruption. The disaster-sensitive sector interaction is the cleanest: the combined group of disaster-sensitive firms that experienced a hurricane defaults at 31.8%, with the hurricane effect (approximately +10.7 points) and the sensitive-sector effect (approximately +8.4 points) combining approximately additively.

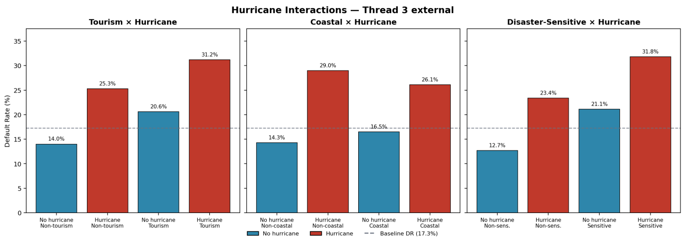

Figure 8. Hurricane interaction effects with tourism sector (left), coastal geography (center), and disaster-sensitive sector classification (right). The dashed reference line marks the 17.3% overall training-set default rate.

From this analysis, nine feature families are constructed. The baseline exposure flags are `tourism_industry_flag`, `coastal_state_flag`, and `disaster_sensitive_sector_flag`. The disaster-type indicators are `fema_hurricane_12m`, `fema_fire_12m`, and `fema_severe_storm_12m`. The interaction features are `tourism_hurricane`, `coastal_hurricane`, and `disaster_sensitive_hurricane`. Flood, drought, tornado, and earthquake flags are treated as report-only because their direction is reversed and their interpretation is dominated by selection rather than by a clear risk signal.

4.4. Feature Engineering Summary

Table 5 consolidates the full set of engineered and derived features that advance to the final model, grouped by source. The top panel lists the nine variables engineered directly from SBA application fields through log transformations, financial ratio derivations, and binary proxies. The bottom panel lists the thirteen EDA-motivated features that capture conditional risk signals from external data and their interactions with borrower characteristics.

A note on the interaction features. The construction × HPI, supply-chain × inflation, and disaster-sensitive sector × hurricane cells reported above show deltas that are approximately additive across the main effects. We do not claim that these interactions carry strong incremental signal beyond the main effects in the aggregate. We retain them because they isolate narrow subpopulations (construction firms during housing downturns, supply-chain firms under inflation, tourism and disaster-sensitive sectors after hurricanes) where the economic story is unambiguous and the resulting flag is interpretable to a loan officer. The interaction features therefore serve an audit-trail purpose rather than a profit-lift purpose. Their aggregate profit contribution is examined directly in Section 6.3.

Table 5. Feature engineering summary. SBA-derived features (top panel) and EDA-motivated external and interaction features (bottom panel).

Channel	Rationale	Feature
	SBA-derived features	
Loan amount	Log-transform to compress right tail	`log_GrAppv`
Firm size	Log-transform to compress right tail	`log_NoEmp`
Guarantee structure	SBA guarantee share of gross approval	`Portion`
Job impact	Jobs created plus retained (sum of CreateJob, RetainedJob)	`– (dropped)`
Employment efficiency	Jobs per dollar of gross approval	`jobs_per_dollar`
Collateral indicator	Term $\geq$ 240 months $\Rightarrow$ real estate	`RealEstate`
Lender proximity	Borrower and bank in same state	`same_state_bank`
	EDA-motivated external and interaction features	
Borrower fragility	Micro firms +inflation: +12.7pp DR	`infl_micro_biz`
Cost pressure	Oil shock coincides with broad DR rise	`oil_shock_supply`
Local demand	Selection effect, tighter underwriting	`income_decl_consumer`
Credit cycle	Bank failure states: lower DR (selection)	`recent_bank_failure`
Housing stress	RE + HPI fall: 35.9% DR	`real_estate_industry_hpi`
Housing stress	Construction + HPI fall: 26.5% DR	`construction_hpi`
Cost pressure	Supply-chain + inflation: 28.6% DR	`supply_chain_infl`
Sector exposure	Tourism DR 22.4% vs 15.7% baseline	`tourism_flag`
Geography	Coastal DR 18.4% vs 15.7% non-coast	`coastal_flag`
Sector exposure	Sensitive DR 23.0% vs 14.1%	`dsast_sensitive`
Hurricane	Exposure: +11.3pp DR delta	`fema_hurricane`
Fire	Exposure: +6.5pp DR delta	`fema_fire`
Storm	Exposure: +5.4pp DR delta	`fema_storm`
Tourism x hurricane	Combined: 31.2% DR	`tourism_hurricane`
Coastal x hurricane	Inland hurricane = highest single DR	`coastal_hurricane`
Sensitive x hurricane	Combined: 31.8%, nearly additive	`dsast_hurricane`

5. MODEL DEVELOPMENT

5.1. Logistic Regression and Linear Methods

Logistic regression remains the industry standard for credit scoring, partly because of its simplicity and partly because regulators understand it. The model estimates the log-odds of default as a linear combination of predictors:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \sum_{j=1}^{k} \beta_j X_j \tag{2}$$

The coefficients β_j are fitted by minimizing the penalized negative log-likelihood. With L2 regularization, the objective becomes:

$$\min_{\boldsymbol{\beta}} \left\{ -\sum_{i=1}^{n} [y_i \log p_i + (1-y_i)\log(1-p_i)] + \lambda \|\boldsymbol{\beta}\|_2^2 \right\} \tag{3}$$

We implement three configurations with $C = 1/\lambda \in \{0.1, 1.0, 10.0\}$. The solver is `lbfgs`, a quasi-Newton method that approximates the inverse Hessian from a limited history of gradients and parameter updates without ever materializing the $k \times k$ matrix. On our 194-dimensional feature space this is substantially faster than Newton's method and more stable than first-order solvers when features vary in scale; ours do, with standardized continuous variables sitting alongside binary one-hot columns.

Table 6. Algorithm 1. Logistic regression with L2 regularization.

Algorithm 1. Logistic regression with L2 regularization (Ridge)

Input: Training data $\{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$, penalty C, max iterations T
initialize $\boldsymbol{\beta} \leftarrow \mathbf{0}$
for $t = 1$ **to** T **do**
  $p_i \leftarrow \sigma(\mathbf{x}_i^\top \boldsymbol{\beta}) = 1/(1+e^{-\mathbf{x}_i^\top \boldsymbol{\beta}}) \quad \forall i$
  $\nabla\mathcal{L} \leftarrow -\sum_i \left[y_i(1-p_i)\mathbf{x}_i - (1-y_i)p_i\mathbf{x}_i\right] + 2\lambda\boldsymbol{\beta}$
  update $\boldsymbol{\beta}$ via L-BFGS two-loop recursion approximating $(\nabla^2\mathcal{L})^{-1}\nabla\mathcal{L}$
  if $\|\nabla\mathcal{L}\| < 10^{-5}$ **then break**
end for
for $\tau \in \{0.01, 0.015, \ldots, 0.60\}$ **do**
  $\Pi(\tau) \leftarrow \sum_i \Pi(\mathbf{1}[\hat{p}_i \le \tau], y_i, a_i)$ (Eq. 1)
end for
$\tau^* \leftarrow \arg\max_\tau \Pi(\tau)$
Output: coefficients $\boldsymbol{\beta}$, threshold τ^*, predicted probabilities $\hat{p}_i$

We also fit Linear Discriminant Analysis (shared covariance $\boldsymbol{\Sigma}$, `lsqr` solver with auto-shrinkage) and Quadratic Discriminant Analysis (class-specific $\boldsymbol{\Sigma}_k$, regularization `reg_param` $= 0.2$). Neither approaches the performance of logistic regression on this data. LDA's shared-covariance assumption is too restrictive when categorical variables have fundamentally different variance structures across default and non-default classes. QDA's unconstrained covariance estimation breaks down in 194 dimensions.

5.2. Random Forest

A single decision tree is easy to explain but prone to overfitting. The random forest remedy is to grow many trees on bootstrap samples, each tree seeing a random subset of features at every split, and average their predictions. The diversity across trees cancels the variance inflation of any individual tree.

The Gini impurity at node D is $G(D) = 1 - \sum_c p_c^2$, where p_c is the proportion of class c in the node. At each split, the algorithm searches over all features j and candidate split points s, selecting the pair (j^*, s^*) that maximizes the impurity reduction:

$$\Delta G = G(D) - \frac{|D_L|}{|D|}G(D_L) - \frac{|D_R|}{|D|}G(D_R) \tag{4}$$

After growing B trees $\{T_b\}_{b=1}^{B}$, the predicted default probability for a new observation $\mathbf{x}$ is the average of the leaf-level default proportions across all trees: $\hat{p}(\mathbf{x}) = \frac{1}{B}\sum_b T_b(\mathbf{x})$.

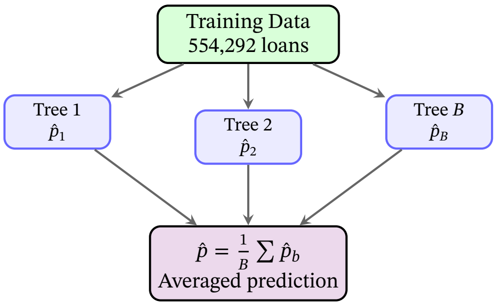

Figure 9. Random Forest: bootstrap samples produce diverse trees whose predictions are averaged.

An important implementation choice concerns `max_features`. The conventional recommendation for classification is $\sqrt{m}$, which balances tree diversity against individual tree strength. We use `max_features = None` (all features at every split) so that each tree can always see the sparse engineered interactions, which fire on roughly 5–10% of observations and would be invisible to a $\sqrt{44} \approx 7$-feature subsample at many nodes. The cost of this choice is reduced tree diversity, and Random Forest is the model family that depends most heavily on diversity for variance reduction; we report the resulting performance in Table 11 and discuss its position in the benchmark there.

Table 7. Algorithm 2. Random Forest.

Algorithm 2. Random Forest

Input: Training set $\{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$, trees B, max depth d, min leaf ℓ
for $b = 1$ **to** B **do**
  draw bootstrap sample $\mathcal{D}_b$ of size n from $\mathcal{D}$ with replacement
  grow tree T_b on $\mathcal{D}_b$:
    at each node, select split (j^*, s^*) maximizing ΔG (Eq. 4) over all features
    stop if node size $< \ell$ or depth $= d$
    assign leaf prediction as mean y in leaf
end for
$\hat{p}(\mathbf{x}) \leftarrow \frac{1}{B}\sum_b T_b(\mathbf{x})$
select τ^* maximizing validation profit
Output: ensemble $\{T_b\}$, threshold τ^*

5.3. Gradient Boosting Methods

5.3.1. XGBoost

Gradient boosting builds an additive model sequentially. At iteration t, a new tree f_t is trained to predict the negative gradient of the loss evaluated at the current ensemble prediction F_{t-1}. The ensemble is updated as $F_t = F_{t-1} + \eta \cdot f_t$, where the learning rate η shrinks each tree's contribution to prevent overfitting.

XGBoost refines this procedure with a second-order approximation. Rather than fitting to the gradient alone, it uses both the gradient g_i and the Hessian h_i of the loss, which allows the tree to account for the curvature of the loss surface at each point:

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^{n}\left[g_i f_t(\mathbf{x}_i) + \frac{1}{2}h_i f_t^2(\mathbf{x}_i)\right] + \gamma T + \frac{1}{2}\lambda\sum_{j=1}^{T} w_j^2 \tag{5}$$

Here T is the number of leaves in tree f_t, w_j is the weight assigned to leaf j, γ penalizes adding leaves, and λ is an L2 penalty on leaf weights. For a fixed tree structure, the optimal leaf weight has a closed form: $w_j^* = -(\sum_{i\in I_j} g_i)/(\sum_{i\in I_j} h_i + \lambda)$. Substituting back yields the split gain formula used in Algorithm 3.

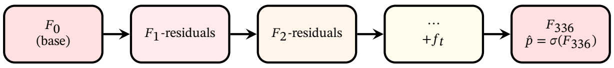

Figure 10. Gradient boosting: each tree corrects residuals from the current ensemble.

Table 8. Algorithm 3. Gradient boosting (XGBoost).

Algorithm 3. Gradient boosting with second-order approximation (XGBoost)

Input: Training data $\mathcal{D}$, learning rate η, trees M, max depth d, γ, λ
$\hat{y}_i^{(0)} \leftarrow \log(\bar{y}/(1-\bar{y}))$ (base log-odds)
for $m = 1$ **to** M **do**
  $g_i \leftarrow p_i^{(m-1)} - y_i, \quad h_i \leftarrow p_i^{(m-1)}(1 - p_i^{(m-1)})$
  grow tree f_m to depth d by maximizing:
    $\text{Gain} = \frac{1}{2}\left[\frac{(\sum_L g)^2}{\sum_L h+\lambda} + \frac{(\sum_R g)^2}{\sum_R h+\lambda} - \frac{(\sum_{L+R} g)^2}{\sum_{L+R} h+\lambda}\right] - \gamma$
  assign $w_j^* = -(\sum_{I_j} g)/(\sum_{I_j} h + \lambda)$ to each leaf j
  $\hat{y}_i^{(m)} \leftarrow \hat{y}_i^{(m-1)} + \eta \cdot f_m(\mathbf{x}_i)$
end for
$\hat{p}_i \leftarrow 1/(1 + e^{-\hat{y}_i^{(M)}})$
select τ^* maximizing validation profit
Output: model F_M, threshold τ^*

5.3.2. *LightGBM*

LightGBM departs from conventional gradient boosting in three ways. First, it grows trees leaf-wise rather than level-wise. At each step, the algorithm finds the single leaf whose split would produce the largest loss reduction and splits only that leaf. This is more aggressive than splitting every node at a given depth, but it achieves lower training loss with fewer total leaves. Our 554,292-row training set provides enough data that overfitting from leaf-wise growth is not a concern.

Second, LightGBM applies Gradient-based One-Sided Sampling (GOSS). Instances with small gradients are already well-predicted by the current ensemble, so excluding a random subset of them from split finding has minimal effect on the selected split point while reducing the effective data size. Instances with large gradients—the ones the model currently struggles with—are always retained.

Third, Exclusive Feature Bundling groups one-hot encoded features that rarely take non-zero values simultaneously. A loan belongs to only one state and one NAICS sector, so the 50 state dummies and 20 sector dummies are largely mutually exclusive. EFB compresses these without losing split-point accuracy.

Table 9. Algorithm 4. LightGBM with leaf-wise growth.

Algorithm 4. LightGBM (leaf-wise growth, histogram-based boosting)

Input: Data $\mathcal{D}$, learning rate η, estimators M, num_leaves L, min_child s
apply EFB: bundle mutually exclusive feature groups
$\hat{y}_i^{(0)} \leftarrow \log(\bar{y}/(1-\bar{y}))$
for $m = 1$ **to** M **do**:
  initialize tree with root node
  for $\ell = 1$ **to** $L - 1$ **do**: ▷ leaf-wise: split leaf with largest gain
    among current leaves, find j^* with maximum split gain
    if $|\text{leaf } j^*| < 2s$ **then break**
    split leaf j^* into two children; update leaf weights
  end for
  $\hat{y}_i^{(m)} \leftarrow \hat{y}_i^{(m-1)} + \eta \cdot f_m(\mathbf{x}_i)$
end for
$\hat{p}_i \leftarrow \sigma(\hat{y}_i^{(M)})$; **select** τ^* maximizing Π_{total} on validation
Output: ensemble $\{f_m\}$, threshold τ^*

5.4. Multi-Layer Perceptron

Our neural network is a feedforward multi-layer perceptron with L hidden layers. Forward propagation at layer ℓ computes $\mathbf{a}_\ell = g(\mathbf{W}_\ell \mathbf{a}_{\ell-1} + \mathbf{b}_\ell)$, where g is the activation function. The output layer applies a logistic sigmoid to produce $\hat{p}$. The network minimizes regularized binary cross-entropy:

$$\mathcal{L} = -\frac{1}{n}\sum_{i=1}^{n}[y_i \log \hat{p}_i + (1-y_i)\log(1-\hat{p}_i)] + \frac{\alpha}{2}\|\mathbf{W}\|_2^2 \quad (6)$$

We compare two activation functions throughout. ReLU, $g(z) = \max(0, z)$, has become the default for feedforward networks because its gradient is either 0 or 1, never a vanishing fractional value, so gradients propagate through many layers without decay. ReLU also induces sparsity: neurons that receive negative input produce zero output and zero gradient, effectively dropping out of the computation. Tanh, $g(z) = (e^z - e^{-z})/(e^z + e^{-z})$, is zero-centered, which can accelerate convergence when inputs are standardized, but saturates to ± 1 for large $|z|$, slowing gradient flow in deeper architectures.

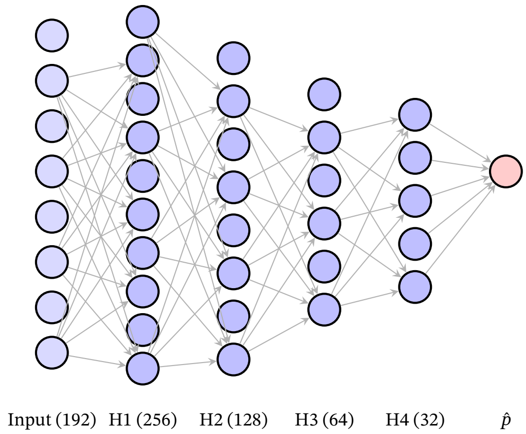

Figure 11. MLP architecture with four hidden layers. Input (192 features) compresses through 256-128-64-32 units to a single sigmoid output.

We use the Adam optimizer rather than plain stochastic gradient descent. Adam maintains exponentially decaying averages of past gradients (first moment) and squared gradients (second moment), using these to compute adaptive per-parameter learning rates. The update takes the form:

$$m_t = \beta_1 m_{t-1} + (1-\beta_1)\nabla\mathcal{L}, \quad v_t = \beta_2 v_{t-1} + (1-\beta_2)(\nabla\mathcal{L})^2 \quad (7)$$

$$\mathbf{W}_{t+1} = \mathbf{W}_t - \eta \cdot \frac{m_t/(1-\beta_1^t)}{\sqrt{v_t/(1-\beta_2^t)} + \epsilon} \quad (8)$$

with $\eta = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$. The bias correction terms $(1-\beta^t)$ account for the fact that m_0 and v_0 are initialized to zero, which would bias the early updates downward without correction.

Early stopping monitors validation loss after each epoch. If the loss fails to improve for ten consecutive epochs, training terminates. We reserve 15% of the training partition for this validation signal. The maximum iteration budget is 500 epochs, though early stopping typically terminates training well before this limit.

5.5. Hyperparameter Tuning

After the benchmark comparison identifies LightGBM as the strongest family, we apply Optuna's Tree-structured Parzen Estimator (TPE) to search for a configuration that improves on the best hand-specified entry. The results of this search, including the final parameters and the optimization trajectory, are presented alongside the benchmark leaderboard in Section 6.2.

Table 10. Algorithm 5. Multi-Layer Perceptron with Adam.

Algorithm 5. Multi-Layer Perceptron with Adam optimization

Input: Training data $\mathcal{D}$, hidden layers $\mathbf{h} = (h_1, \dots, h_L)$, activation g, α, η
initialize $\mathbf{W}_\ell \sim \mathcal{N}(0, \sqrt{2/\text{fan\_in}})$ (He initialization)
initialize $\mathbf{m}_\ell \leftarrow \mathbf{0}$, $\mathbf{v}_\ell \leftarrow \mathbf{0}$ for all layers
for $t = 1$ **to** 500 **do**
  forward: $\mathbf{a}_0 \leftarrow \mathbf{x}$; for $\ell = 1..L$: $\mathbf{a}_\ell \leftarrow g(\mathbf{W}_\ell \mathbf{a}_{\ell-1} + \mathbf{b}_\ell)$
  $\hat{p} \leftarrow \sigma(\mathbf{W}_{L+1}\mathbf{a}_L + \mathbf{b}_{L+1})$
  compute $\mathcal{L}$ and $\nabla_{\mathbf{W}}\mathcal{L}$ via backpropagation
  update all $\mathbf{W}_\ell$ using Adam (Eq. 7)
  if validation loss not improved for 10 epochs **then break**
end for
select τ^* maximizing validation profit
Output: weights $\{\mathbf{W}_\ell\}$, threshold τ^*

6. RESULTS AND PROFIT-BASED APPROVAL RULE

6.1. Model Comparison

Table 11 presents the validation performance of 21 benchmark model configurations on the cleaned 44-feature set, ranked by net profit. Every model was fitted on the same 554,292 training loans and evaluated on 138,573 validation loans. Thresholds were individually tuned to maximize profit. The 44-feature set reflects the removal of 10 redundant variables identified through multicollinearity analysis (see Section 3.5). Appendix A provides the full 21-model benchmark table together with the separate Optuna LightGBM follow-up; Appendix B provides visual comparisons of model performance across families.

Table 11. Validation leaderboard — 44 cleaned features, profit-tuned thresholds. Top configurations from the 21-model benchmark board plus the Optuna-tuned LightGBM follow-up; the complete 22-row table is in Appendix A.

Model	Profit (M$)	AUC	Brier	Time (s)
LGB Optuna (n418 nl194)*	**1,096.4**	0.9827	0.0335	–
LGB n300 nl127	1,078.6	0.9814	0.0351	12
LGB n300 nl63	1,072.3	0.9803	0.0365	8
Bagging n100	1,062.3	0.9774	0.0399	54
XGB n300 d10	1,058.1	0.9793	0.0367	6
LGB n200 nl31	1,055.1	0.9772	0.0399	4
XGB n200 d8	1,045.5	0.9770	0.0393	3
HGB i300 d63	1,044.4	0.9776	0.0388	27
HGB i200 d31	1,032.2	0.9745	0.0420	13
AdaBoost n100	1,020.0	0.9712	0.1561	132
MLP 256x128x64x32	978.5	0.9595	0.0514	91
RF n100 d16 l25	947.2	0.9541	0.0592	8
RF n200 d16 l50	934.8	0.9520	0.0608	16
Ridge C=0.1	790.8	0.8745	0.0899	13
LDA	769.3	0.8606	0.0961	7

**Recommended model (Optuna-tuned, 30 TPE trials). LGB n300 nl127 is the best pre-specified benchmark entry at $1,078.6M. Optuna params: n_est=418, nl=194, lr=0.064, min_child=85, subsample=0.50, colsample=0.82, reg_alpha=0.001, reg_lambda=0.017. Training set: 554,292 loans; validation: 138,573 loans.*

LightGBM. The Optuna-tuned LightGBM leads the board at $1,096.4M (AUC 0.9827, $\tau = 0.1488$, 77.8% approval, 1.09% DR, 4.70% ROI). A 30-trial TPE search found this configuration (n_est=418, nl=194, lr=0.064, subsample=0.50), which improves on the best hand-specified entry (n300 nl127, $1,078.6M) by $17.8M. The multi-seed validation confirms stability: $1,091.7M ± $4.1M across three seeds, with the Optuna-tuned LightGBM winning every seed. The second manual configuration (n300 nl63, $1,072.3M) and the lighter variant (n200 nl31, $1,055.1M at 4s) show that even compact LightGBM configurations remain competitive.

Bagging. The bagged decision-tree ensemble ranks third ($1,062.3M, AUC 0.9774). Its performance confirms that a relatively simple ensemble of trees already captures much of the non-linear signal in the SBA application file. Relative to LightGBM, however, it gives up about $16M of validation profit and a small amount of ranking quality. In this project, bagging serves as a strong robustness benchmark rather than the best operating model.

XGBoost. The two XGBoost configurations ($1,058.1M and $1,045.5M) are competitive with the best manual LightGBM on AUC (0.9793 vs 0.9814) but trail by $20.5M on profit. The gap is not from ranking quality—the AUC difference is modest—but from the threshold-optimized decision: XGBoost's optimal τ is 0.1439, approving 77.1% of applicants, while the best manual LightGBM reaches a higher profit at its profit-optimal threshold.

HistGradientBoosting. The two HGB configurations ($1,044.4M and $1,032.2M) produce AUC values comparable to XGBoost (0.9776 vs 0.9793) but trail it on profit and train more slowly. On this dataset, HGB remains a credible tree-based benchmark, but it does not offer a profit or runtime advantage over LightGBM.

AdaBoost. AdaBoost gained from feature cleaning, rising to $1,020.0M. The exponential loss function is sensitive to noisy features: each misclassified observation receives exponentially increasing weight, so irrelevant variables that create spurious boundaries cause the algorithm to chase noise. Removing ten collinear features let the ensemble concentrate on genuine risk patterns. AdaBoost's Brier score (0.1561) is more than four times LightGBM's (0.0335), indicating poorly calibrated probabilities despite reasonable ranking (AUC 0.9712). The exponential loss pushes probability estimates toward 0 and 1, overweighting hard cases at the expense of calibration.

Multi-Layer Perceptron. The three MLP architectures cluster between $970.9M and $978.5M. The deepest configuration (256-128-64-32, ReLU, Adam) achieves the highest profit at $978.5M with AUC 0.9595 and $\tau = 0.0992$. The four-layer architecture outperforms the three-layer variant by $6.2M, suggesting diminishing returns from additional depth on this tabular dataset. MLPs require dense one-hot encoding (183 columns), z-score standardization, and substantially longer training (91–97 seconds). While they outperform all linear models by over $187M, they trail the best tree ensemble by $100.1M. The gap reflects a structural limitation: feedforward networks with ReLU activations learn smooth, continuous decision boundaries, whereas SBA credit risk exhibits threshold effects (loan term above 240 months, employee count below 5, HPI declining) that are naturally captured by axis-aligned tree splits.

Random Forest. The three RF configurations cluster between $934.8M and $947.2M, well below the boosted-tree group. Their AUC values around 0.95 show that the models still rank risk meaningfully, but their approval decisions are less favorable near the operating threshold than those of LightGBM, XGBoost, or Bagging. In this setting, simple bootstrap averaging is not enough to match the finer probability ranking produced by the boosting models.

Logistic Regression (Ridge). The three ℓ_2-regularized logistic regressions ($790.8M–$791.4M) are nearly indistinguishable, confirming that the regularization strength C has minimal effect on this dataset when the feature count is moderate and the training set is large. All three achieve AUC around 0.875 and approve roughly 72–73% of applicants. The $287M gap to LightGBM is the cost of linearity: logistic regression cannot capture interactions between loan term and industry, between guarantee share and geography, or between external macro conditions and borrower characteristics without explicit interaction terms. ℓ_2 regularization stabilizes coefficient estimates in the presence of residual collinearity but cannot create the non-linear structure that the data demands.

LDA and QDA. Linear Discriminant Analysis achieves $769.3M with AUC 0.8606. Quadratic Discriminant Analysis is materially weaker, landing between $693.0M and $697.4M with AUC between 0.789 and 0.794. QDA's quadratic decision boundaries should, in principle, capture more non-linear structure than LDA's linear hyperplane. In practice, QDA estimates a separate covariance matrix for each class (44×44 parameters per class), and with 23,935 defaults in the training set, the minority-class covariance is poorly estimated. The regularization parameter (r=0.1–0.2) shrinks the class covariances toward the pooled estimate, improving stability but at the cost of reverting toward the LDA solution. QDA's Brier scores (0.174–0.187) are the worst of any model family, making its probability estimates unreliable for threshold-based decision rules.

Summary. The comparison board points to a clear pattern (see also Appendices A–B for the complete numerical tables and performance charts). Tree ensembles occupy the top of the ranking, and the boosted variants outperform bagging and random forests. The profit objective also separates models that look similar on AUC alone: XGBoost nearly matches LightGBM's ranking quality but trails on profit because its thresholded approval rule is slightly less favorable near the operating point. Within the fixed 21-model benchmark board, LightGBM n300 nl127 is the strongest pre-specified entry ($1,078.6M). The Optuna-tuned extension lifts this to $1,096.4M and is the recommended model.

6.2. Optuna-Tuned LightGBM: Results and Final Selection

After the benchmark comparison identified LightGBM as the leading family, we ran a 30-trial Optuna search using the Tree-structured Parzen Estimator (TPE) on the cleaned 44-feature matrix. TPE models the distribution of hyperparameter values separately for good and bad trials: it fits a kernel density estimator $\ell(\theta)$ to the top γ quantile of observations and $g(\theta)$ to the remainder, then samples the next trial by maximizing $\ell(\theta)/g(\theta)$. Each trial trains on the 554,292-row training set, predicts on the 138,573-row validation set, and returns the profit-maximizing threshold value as the objective.

The best trial reaches $1,096.4M with parameters $n_{\text{est}} = 418$, $n_{\text{leaves}} = 194$, `lr` = 0.064, `min_child` = 85, `subsample` = 0.50, `colsample` = 0.82, `reg_alpha` = 0.001, `reg_lambda` = 0.017. The aggressive subsample (0.50) introduces substantial per-tree diversity, and the large number of leaves (194) with low ℓ_2 regularization allows deep trees that capture fine-grained risk patterns. Fig. 12 shows the 30-trial trajectory; the best trial is the last (trial 29).

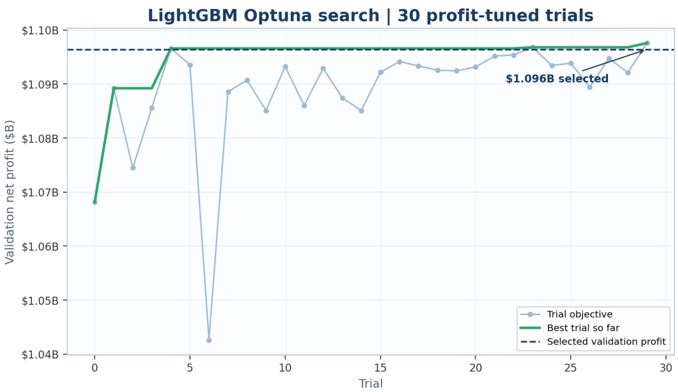

Figure 12. LightGBM Optuna optimization history (30 trials, TPE sampler). The best trial reaches $1,096.4M; the manually specified benchmark LightGBM (n300 nl127, $1,078.6M) is shown for comparison.

The Optuna-tuned configuration reaches $1,096.4M, a $17.8M improvement over the best manually specified entry on the benchmark board. The gains come from two directions: more trees (418 vs 300) with larger leaves (194 vs 127) allow deeper risk stratification, and aggressive subsampling (0.50 vs 0.80) increases per-tree diversity. Since this configuration produces the highest validation profit among all methods tested—and the gap exceeds the multi-seed standard deviation of $4.1M—we select it as the recommended model. The benchmark leaderboard retains its fixed-configuration comparison value; the Optuna result extends it.

6.3. Return on Approved Exposure

While net profit measures total dollar return, a lender also cares about the *rate* of return per dollar lent. We define ROI on approved exposure as net profit divided by the total disbursement amount of approved loans: $\text{ROI} = \Pi_{\text{net}} / \sum_{i:\hat{p}_i \le \tau} a_i$. This metric penalizes approval decisions that generate profit in absolute terms but at low marginal efficiency.

Table 12 reports ROI for the top ten models alongside their approval rates. LGB Optuna achieves the highest ROI at 4.70%, followed by Bagging n100 at 4.63%, approving $23.6B in exposure. LGB n200 nl31 (4.61%) and LGB n300 nl63 (4.58%) are competitive on rate-of-return despite trailing on total profit, because their profit-optimal thresholds produce approval rates close to LightGBM's. The pattern is monotonic across families: tree models cluster between 4.46% and 4.63%, MLPs around 4.28%, random forests at 4.10%–4.13%, and linear models at 3.32%–3.34%.

Table 12. ROI on approved exposure for top models.

Model	ROI	Approval	Exp. ($B)
LGB Optuna (n418 nl194)	4.70%	77.8%	22.9
LGB n300 nl63	4.58%	77.9%	22.9
LGB n200 nl31	4.61%	76.4%	22.6
Bagging n100	4.63%	76.9%	22.8
XGB n200 d8	4.52%	76.9%	22.6
XGB n300 d10	4.57%	77.1%	22.9
HGB i300 d63	4.47%	77.9%	22.7
HGB i200 d31	4.48%	76.5%	22.5
AdaBoost n100	4.46%	75.9%	22.4
MLP 256x128x64x32	4.28%	75.3%	22.4

ROI = net profit / approved exposure. Exposure is the sum of DisbursementGross for approved loans.

The ROI ranking is compressed relative to the profit ranking. The best model earns only 14 basis points more than the tenth model on return per dollar, yet $87.4M more in total profit. This is because marginal loans near the approval threshold carry near-zero expected profit: approving them increases exposure without proportionally increasing net profit, diluting ROI while adding to total profit. Second, the negative opportunity cost of denying good loans is invisible to ROI but directly subtracts from profit. A model that is too conservative (low τ) achieves high ROI on the loans it does approve, but leaves profit on the table by rejecting creditworthy applicants. The profit-maximizing threshold balances these forces; the ROI values in Table 12 describe the outcome of that balance, not a separate optimization target. The profit-versus-AUC scatter plot in Appendix B (Fig. 21) already illustrates this tradeoff visually: models that score highest on AUC are not necessarily those that earn the most profit, and the ROI ranking is similarly compressed.

6.4. Baseline SBA-only versus External Context

We compare the Optuna-tuned LightGBM trained on three feature sets under the identical 80/20 train/valid split: SBA-only (16 features: 9 baseline numeric + 7 categorical), SBA+external without interaction terms (31 features), and the full 44-feature set with all 13 engineered interactions.

Table 13. SBA-only versus SBA-plus-external: validation performance.

Feature set	Profit	AUC	ROI
SBA-only (16)	$1,087.9M	0.9809	–
SBA+external (31)	$1,093.4M	0.9825	–
SBA+ext+interact (44)	$1,096.4M	0.9827	4.70%

On the shared split, external variables raise validation profit from $1,087.9M in the SBA-only specification to $1,096.4M in the full 44-feature specification, a gain of $8.5M (0.78%). The interaction terms do not add aggregate profit on their own: the 31-feature specification without them reaches $1,093.4M, slightly below the full set. Their role is interpretive and local: they isolate narrow settings where the economic mechanism is clear even if the aggregate contribution is small.

The same comparison also clarifies what external data does not do. The SBA-only model already ranks risk extremely well (AUC 0.9809), so the external lift is incremental rather than transformational. Its value is in sharpening the approval-time economic context while still improving profit on the common validation split. As a point of reference, a separate 30-trial Optuna search on the full 44-feature specification reaches $1,096.4M, a further $17.8M above the manually specified n300 nl127 (Table 11).

6.5. Held-Out Test Set Confirmation

The 80/20 split reserves 173,217 loans (20%) as a held-out test partition. We confirm the winning model's performance by retraining the Optuna-tuned LightGBM on the full train+valid set (692,865 loans) and evaluating on the 173,217 test loans at the selected threshold $\tau = 0.1488$.

Table 14. LGB Optuna (n418 nl194): held-out test evaluation at the validation-selected cutoff $\tau = 0.1488$.

Metric	Value
AUC	0.9826
Brier Score	0.0339
Approval rate	77.7%
Approved default rate	1.11%
Profit	$1,358.3M
Profit lift over approve-all	$785.4M
ROI on approved exposure	4.67%

Model retrained on the full 692,865-loan train+validation sample. Threshold $\tau = 0.1488$ selected during model comparison. Approval-outcome counts on the test set: correctly approved = 133,077, good loans denied = 10,223, defaults approved = 1,498, defaults denied = 28,419.

The test set performance confirms that the rule generalizes: AUC of 0.9826 versus 0.9827 on validation, Brier of 0.0339, and profit of $1,358.3M at the validation-selected threshold $\tau = 0.1488$. The confusion matrix on the 173,217 test loans is TN=133,077 (correctly approved), FP=10,223 (good loans denied), FN=1,498 (defaults approved), TP=28,419 (defaults denied), yielding a 77.7% approval rate and a 1.11% approved default rate.

6.6. The Profit Lift Curve

Fig. 13 translates model predictions into an approval policy. It sorts all validation-set loans by predicted probability of default, from lowest to highest, and plots cumulative net profit as the lender progressively approves more applicants.

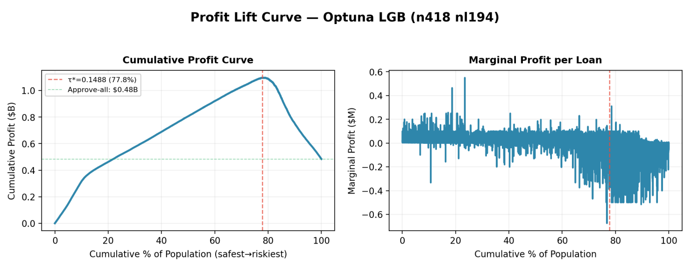

Figure 13. Profit lift curve. Left: cumulative net profit versus approval rate. Red dashed line marks the optimum at 77.8% approval ($1,096.4M). Green dotted line shows approve-all profit ($483.9M). Right: marginal profit per loan beyond the optimum turns negative.

The curve rises steadily through the first eight deciles. Profit is already positive after the safest quarter of applicants is approved, and cumulative profit continues to increase until the approval rate reaches roughly 78%, where it peaks at $1,096.4M. Beyond that point, the incremental approvals are disproportionately concentrated among high-risk borrowers, so cumulative profit declines even though the approval rate rises.

The difference between the profit maximum and the approve-all endpoint is $612.5M. Under the study's payoff assumptions, that amount is the economic value of screening out the riskiest tail of the portfolio rather than lending indiscriminately.

6.7. Gains and Lift

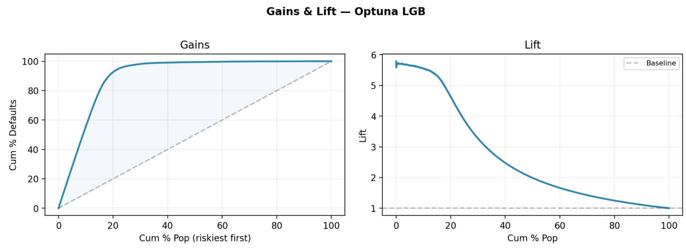

Figure 14. Gains chart (left): the first 50% of applicants contain under 5% of all defaults. Lift chart (right): the cumulative default rate among approved applicants remains below the baseline rate through the first eight deciles.

Fig. 14 shows the gains and lift charts. The gains curve reveals that the first 25% of applicants, when ordered by predicted PD, contain fewer than 1% of all defaults. Half of all defaults are concentrated in the riskiest 15% of the pool. The lift curve confirms that the cumulative default rate of approved applicants stays below the baseline default rate until the approval depth exceeds roughly 80%.

6.8. Decile Analysis

To examine how risk and profit concentrate across the applicant distribution, we sort all 138,573 validation loans by predicted probability of default from lowest to highest and divide them into ten equal-sized groups (deciles). Each decile contains approximately 13,857 loans.

Table 15. Decile risk profile, sorted by increasing predicted default probability.

D	PD Range	Loans	DR	Cum. Profit (M$)
D1	0.0000–0.0003	13,857	<0.01%	316.0
D2	0.0003–0.0012	13,857	0.09%	462.4
D3	0.0012–0.0019	13,857	0.14%	574.1
D4	0.0019–0.0031	13,857	0.28%	689.0
D5	0.0031–0.0053	13,857	0.48%	806.1
D6	0.0053–0.0110	13,857	0.62%	920.6
D7	0.0110–0.0374	13,857	1.59%	1,027.1
D8	0.0374–0.2435	13,857	9.51%	**1,087.6**
D9	0.2435–0.8970	13,857	64.03%	752.1
D10	0.8971–0.9985	13,860	95.94%	483.9

Deciles contain approximately 13,857 loans each. D8 is the last profitable decile, peaking at $1,087.6M cumulative profit before D9 and D10 reverse the gains.

The earliest deciles contain extremely low-risk borrowers, while the last two deciles concentrate most of the expected losses. D8 is the last decile whose approval still adds to cumulative profit; D9 and D10 together erase over $600M accumulated in the safer tiers.

6.9. The Approval Rule

Thresholds between 0.12 and 0.24 keep validation profit within $12M of the optimum, so the approval rule is not fragile to small calibration drift.

Operationally, the recommended rule is to use the Optuna-tuned LightGBM and approve applicants whose predicted probability of default does not exceed the profit-maximizing threshold $\tau = 0.1488$. On the validation set, the Optuna model yields $1,096.4M in net profit.

Table 16. Profit sensitivity to threshold choice for the recommended Optuna-tuned LightGBM model.

PD τ	$1-\tau$	Approval	Profit (M$)
0.10	0.90	75.9%	1,083
0.12	0.88	76.8%	1,090
0.14	0.86	77.5%	1,095
0.1488	**0.8512**	**77.8%**	**1,096.4**
0.20	0.80	79.1%	1,095
0.24	0.76	79.9%	1,088

The manually specified n300 nl127 at $\tau = 0.1934$ earns $1,078.6M and serves as a simpler alternative when retraining frequency matters.

7. FEATURE IMPORTANCE AND MODEL DIAGNOSTICS

7.1. Features That Drive the Final Model

We examine feature importance from two complementary perspectives: LightGBM's built-in gain-based importance and SHAP (SHapley Additive exPlanations) values. Gain-based importance measures how much each feature contributes to loss reduction across all splits in the ensemble. SHAP values decompose individual predictions into additive feature contributions grounded in cooperative game theory.

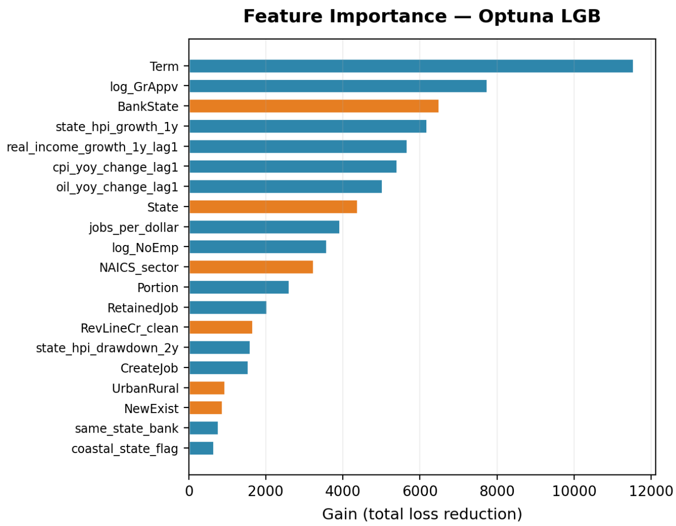

Figure 15. LightGBM feature importance by total gain (loss reduction). Loan amount (log_GrAppv), term, and employment variables dominate the top ranks. External and engineered features provide secondary but consistent signals.

Fig. 15 shows the top 20 features ranked by gain. Term is the single strongest predictor, followed by BankState, log_GrAppv, and external macro indicators such as state-level HPI growth and real-income growth. Among the SBA-derived features, the guarantee ratio (Portion), loan-to-employee efficiency (jobs_per_dollar), and the same-state-bank indicator appear in the upper tier. The external macro variables (HPI trends, inflation flags, FEMA hurricane exposure) contribute meaningful secondary signal as main effects. Consistent with the ablation in Section 13, the engineered interaction features do not appear in the top tier: their predictive contribution is local rather than global, and the aggregate profit lift from external data comes primarily from the main effects rather than from the interaction terms.

Fig. 16 provides a more granular view. High loan amounts (log_GrAppv) push SHAP values negative (lower predicted default), consistent with the EDA finding that larger loans default at lower rates. Short terms push SHAP values positive (higher predicted default), while long terms concentrate on the negative side. Among external features, HPI-negative flags and FEMA hurricane exposure push SHAP values modestly positive, aligning with our external EDA hypotheses. The replacement of raw year variables with external

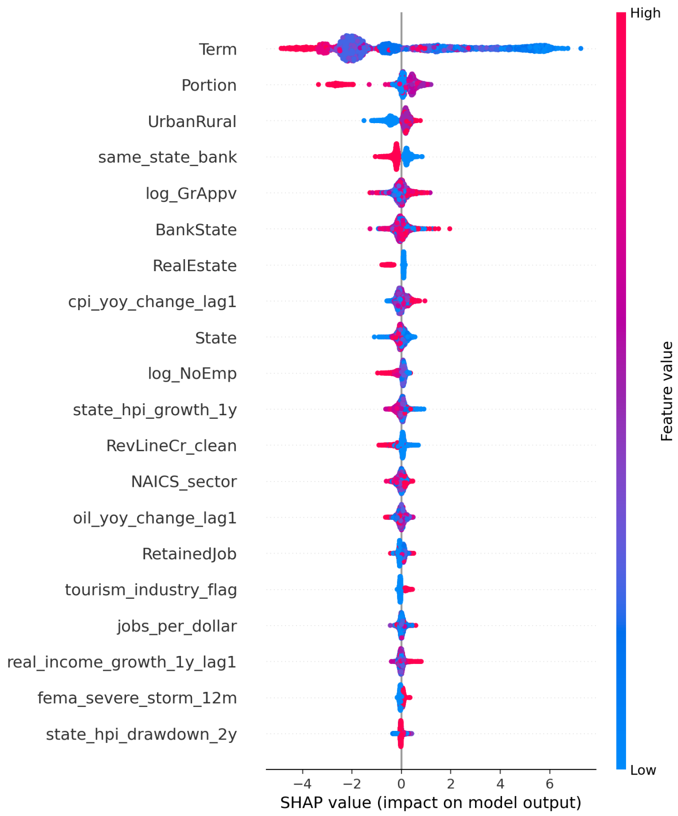

Figure 16. SHAP summary plot for the LightGBM model. Each point represents one validation loan. The horizontal position shows the SHAP value (impact on predicted default probability), with positive values pushing the prediction toward default. Color indicates feature magnitude (red = high). log_GrAppv, Term, and log_NoEmp show the widest dispersion, confirming their dominant role.

macro indicators is visible in the importance rankings: HPI growth, CPI changes, and oil price shifts now carry the cohort-level signal that raw calendar year previously absorbed.

7.2. Interpreting the Engineered Interaction Features

A reader inspecting the feature-importance rankings might conclude that the thirteen engineered interaction features contribute little because none appear among the top 10 by gain or mean SHAP value. That reading is too narrow for variables that were designed to activate only in specific combinations.

The engineered features are not broad main effects. They are interaction flags that encode specific combinations of borrower type and environment. A construction firm is not automatically risky, and a falling HPI is not uniformly risky across all borrowers. The concern arises when both are present, which is what `construction_hpi` captures. The same logic applies to `tourism_hurricane`: it is inactive for most training loans, but it isolates a narrow set of tourism businesses exposed to recent hurricanes, a pattern that no single baseline variable represents cleanly.

This conditional structure changes how the charts should be read. Because the exposure base is narrow, the aggregate gain assigned to an interaction across the full ensemble will be smaller than the gain assigned to a variable such as `GrAppv` that can split almost every observation. Low global importance does not mean the feature is uninformative at the margin. The interactions can also redistribute apparent importance away from their component variables: once the hurricane signal is partly absorbed by `tourism_hurricane`, the standalone disaster flag can appear weaker even though the combined hurricane information has become more precise.

These features encode *when* a borrower becomes riskier, not just whether a sector is risky on average. They save the tree ensembles from reconstructing narrow interactions through deeper split sequences. They also give the lender an auditable explanation for localized denials: a reviewer can identify a construction-housing or tourism-hurricane interaction directly instead of relying on an opaque combination of many tree paths.

7.3. What External Variables Add Beyond the SBA File

The interaction features matter in a narrower way. They identify cases where the same macro condition should not be treated uniformly across the portfolio, such as construction firms during housing declines or tourism firms after hurricanes. Because those cases are relatively sparse, their portfolio-wide profit effect is small even when they are useful for individual credit decisions.

7.4. ROC Curve, KS Statistic, and Calibration

The ROC curves (Fig. 17) show excellent separation between default and non-default classes on both the validation and held-out test samples. The Kolmogorov-Smirnov (KS) statistic on the validation set is 0.8784, meaning the model can separate the cumulative distributions of defaulters and non-defaulters by nearly 88 percentage points at the point of maximum separation. The high KS is consistent with the high AUC but, on its own, also makes the leakage audit in Section 3.2 essential.

Fig. 18 shows the calibration of predicted probabilities on the validation sample. The model's predictions track the observed default rates closely across all probability bins, with only mild over-prediction in the upper range. The validation Brier score of 0.0335 keeps the mean squared probability error low enough that the profit-maximizing threshold remains operationally stable.

The confusion matrix at $\tau = 0.1488$ is reported in Table 19. The model correctly denies 22,757 of 23,935 defaulters, while 1,178 defaults slip through. Among 114,638 performing loans, 106,636 are approved and 8,002 denied. The approved default rate is 1,178 / 107,814 = 1.09%.

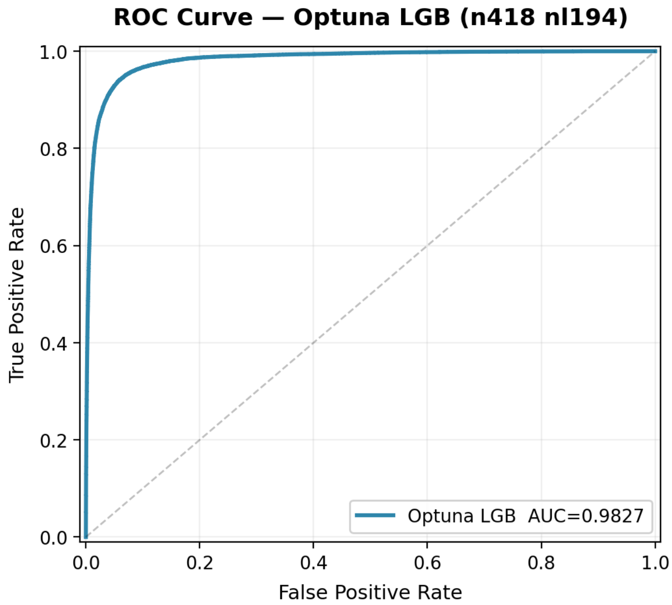

Figure 17. ROC curves for the LightGBM model on the validation set. AUC of 0.9827 confirms strong discriminative ability.

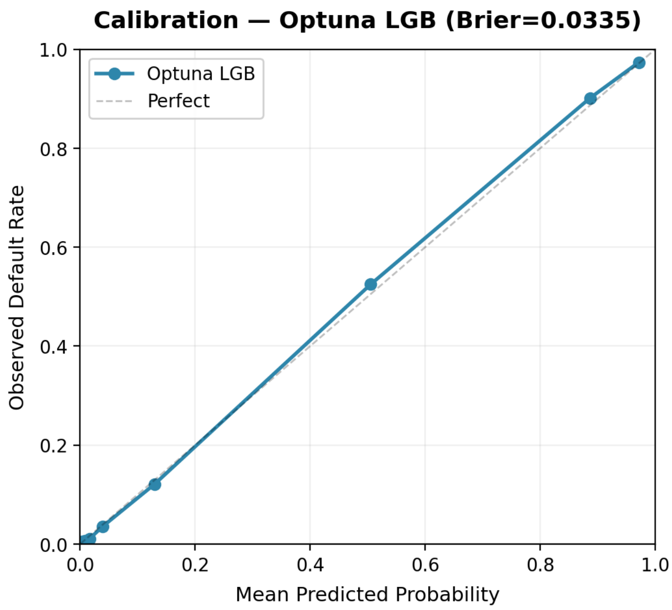

Figure 18. Calibration curve (reliability diagram) for the LightGBM validation predictions. Points above the diagonal indicate under-prediction of default risk; points below indicate over-prediction. The validation Brier score of 0.0335 reflects strong overall calibration.

8. ROBUSTNESS AND VALIDITY CHECKS

8.1. Era-Forward Validation

The main results use an era-stratified random split. To test whether the model transfers across a fundamental regime shift, we move outside the random split entirely. We train exclusively on pre-crisis loans (1987–2007, 782,360 loans, 14.3% DR) and evaluate on crisis loans (2008–2009, 50,401 loans, 29.3% DR). No crisis-era loans appear in training.

The model retains useful discriminative ability on the crisis sample (AUC 0.960), despite the base default rate nearly tripling from the training distribution. The Brier score degrades from 0.035 on in-era validation to 0.067 on the crisis sample, reflecting the calibration shift expected under a large base-rate change. The model was calibrated for a 14% default rate; faced with a population defaulting at 29%, it correctly identifies the riskiest borrowers but underestimates the absolute probability of default. This result supports a limited but meaningful claim: the model's ranking ability transfers across regimes, even though its probability estimates require recalibration when the economic environment changes sharply.

Table 17. Era-forward validation: Optuna-tuned LightGBM trained on pre-crisis and tested on crisis-era loans.

Train era	Test era	Train DR	Test DR	AUC	Brier
Pre (1987–2007)	Crisis (2008–2009)	14.3%	29.3%	0.960	0.067

715,567 training loans; 117,194 test loans. No crisis-era loans appear in training.

8.2. Profit Sensitivity to the Payoff Assumption

The profit rule depends on the assumed 5% gain and 25% loss percentages. To test whether the decision rule is overly sensitive to that assumption, we hold the model fixed and recompute the profit-maximizing threshold under loss rates from 15% to 35%. Table 18 reports the results. As losses become harsher, the optimal cutoff becomes more conservative, which is exactly what the lending logic implies. The resulting validation profit stays within about $55M across the full range, so the business conclusion does not depend on one knife-edge payoff choice.

Table 18. Profit sensitivity to the loss assumption for the recommended Optuna-tuned LightGBM model. The model predictions are held fixed; only the payoff matrix and the profit-maximizing threshold are recalculated.

Gain	Loss	BE PD	Empirical τ	Approval	Profit
+5%	−15%	0.250	0.2331	79.8%	$1,127.7M
+5%	−20%	0.200	0.2034	79.2%	$1,111.2M
+5%	−25%	0.167	0.1488	77.8%	$1,096.4M
+5%	−30%	0.143	0.1488	77.8%	$1,084.6M
+5%	−35%	0.125	0.1488	77.8%	$1,072.7M

8.3. Multi-Seed Stability

The paper's main results use `random_state` = 1. To assess sensitivity to that choice, we retrain four tree-based families (LightGBM, Bagging, XGBoost, HGB) under seeds 1, 42, and 123 using identical hyperparameters on the same split logic. LightGBM places first in every seed with a mean profit of $1,091.7M ± $4.1M. The second and third positions are not strictly ordered: Bagging exceeds XGBoost in seeds 1 and 123, but XGBoost exceeds Bagging by $1.8M in seed 42. HGB is fourth in every run. The practical conclusion is that the selected family (LightGBM) is robust across seeds; the contest for second place between Bagging and XGBoost is within seed noise and either could plausibly be the runner-up.

8.4. Dependence on Core Approval-Time Signals

This check is not about leakage, which was already handled in Section 3, but about concentration of predictive power. We temporarily remove three of the strongest approval-time variables from the final 44-feature set: Term, log_GrAppv, and Portion. Retraining LightGBM on this ablated feature set lowers validation AUC from 0.981 to 0.838 and validation profit from roughly $1,096M to roughly $830M. The drop is large and expected. Loan maturity, loan size, and guarantee structure are among the most economically meaningful signals available at origination. The important point is the residual performance: even after stripping out those dominant features, the model remains well above chance. The ranking signal is therefore not coming from a single disguised shortcut; it is distributed across the remaining borrower, industry, geography, and external-context variables.

8.5. Calibration

The gap between the validation-optimal threshold (0.1488) and the theoretical break-even (0.1667) reflects the model's calibration characteristics under this specific train-validation split. To test calibration directly, we apply Platt (sigmoid) scaling via five-fold cross-validation on an auxiliary stratified split. The calibrated probabilities shift the optimal threshold toward the theoretical break-even but change validation profit by less than $1M, consistent with the finding that the profit surface is flat near its maximum. The main operational cutoff remains 0.1488, selected on the primary validation set used for model comparison.

Table 19. Approval-outcome table for LGB Optuna (n418 nl194) at $\tau = 0.1488$.

	Predicted Approve	Predicted Deny	Total
Actual Paid	106,636 (TN)	8,002 (FP)	114,638
Actual Default	1,178 (FN)	22,757 (TP)	23,935
Total	107,814	30,759	138,573

Approval rate: 77.8%. Approved default rate: 1.09%. TN = correctly approved, FP = good loans wrongly denied, FN = defaults wrongly approved, TP = defaults correctly denied. The model denies 22,757 of 23,935 defaulters (95.1% recall).

9. DISCUSSION

9.1. What the SBA Application Already Captures

The strongest signals in this project come from the original application file. Loan term, loan size, firm size, documentation status, industry, and geography already separate a large share of the risky tail from the rest of the portfolio. That is why the SBA-only LightGBM specification is already strong: it reaches $1,087.9M in validation profit with AUC 0.9809 before any external variables are added. In economic terms, the application file already tells the lender a great deal about borrower fragility, collateral structure, and broad regional exposure at the moment of approval.

9.2. What External Context Added

External variables improved the benchmark, but only at the margin. On the common split, the SBA-plus-external specification reaches $1,096.4M, about $8.5M above the SBA-only version. External context identifies when a given borrower profile becomes riskier: a construction firm is more vulnerable during a housing downturn, and a tourism business is more exposed after a hurricane. The 31-feature model without interaction terms reaches $1,093.4M, below the full 44-feature version. The interaction flags therefore add limited aggregate profit, but they identify specific corners of the book where the risk story is sharper than a main-effect variable alone can provide.

9.3. Model-Selection Rationale

LightGBM was chosen because it gave the best operating trade-off on the benchmark board. Relative to the other tree ensembles, it combined the highest validation profit, the strongest AUC among the pre-specified models, and a short training time. XGBoost came close on ranking quality but produced a less favorable approval rule near the profit-maximizing threshold. Bagging remained competitive and is useful as a robustness benchmark. The winning model converted ranking quality into the most profitable approval rule at a practical approval rate.

9.4. Sample Selection and Censoring

The dataset contains approved SBA loans only. It does not include the applications that lenders rejected. That means the model is trained on a pre-screened population rather than on the full applicant pool. As a result, the recommended cutoff should be interpreted as a rule for ranking and screening within historically approved applicants, not as a fully general reject-or-approve policy for all possible borrowers.

Right-censoring is a second limitation. Loans originated late in the sample have had less time to default than older loans. This is especially relevant for post-2009 originations, which partly explains why their observed default rate is much lower than the crisis-era rate. The held-out test results therefore show that the model transfers well within the observed sample, but they do not remove the broader concern that later-vintage outcomes are less mature.

9.5. Profit Function and External-Data Resolution

The payoff rule is deliberately stylized. We treat an approved performing loan as a 5% gain and an approved default as a 25% loss, regardless of contract rate, collateral recovery, servicing cost, or time to default. That simplification is acceptable for a classroom decision framework, but a real lender would want a loan-level revenue and loss model rather than one common payoff matrix for the entire portfolio.

The external merge also operates at mixed temporal resolutions. FEMA declarations are date-specific, whereas some macro variables are annual or lower-frequency aggregates. That means the external block captures broad approval-time context more reliably than it captures sub-annual local shifts. The external variables are therefore best interpreted as contextual signals rather than as precise real-time state indicators.

9.6. Fairness and Deployment Scope

Several important deployment questions remain outside the scope of this paper. Geographic and industry variables are useful predictors, but they can also correlate with protected characteristics or with historical credit-allocation patterns. We did not run a formal fairness audit across states, industries, or borrower subgroups. We also did not evaluate how the rule would behave under manual review, changing underwriting standards, or a different approval mix. Those are essential steps before any operational deployment.

9.7. One Crisis, One Tuned Configuration

The dataset contains exactly one major macroeconomic regime shift (2008–2009). All era-forward generalization claims in Section 8.1 are based on that single transition. Whether the model's ranking ability would transfer to a future stress event driven by different mechanisms is not testable with the data we have.

10. CONCLUSION

This study compared twenty-one benchmark model configurations for SBA 7(a) loan default prediction under a profit-based lending objective. LightGBM dominated the board, and a 30-trial Optuna search on the winning family lifted validation profit to $1,096.4M. We recommend the Optuna-tuned model for deployment. Relative to approving every loan, the rule adds about $612.5M in validation profit by screening out the riskiest tail of the portfolio.

Three substantive findings follow from the comparison. First, the choice of evaluation metric changes the model ranking: LightGBM and XGBoost are nearly indistinguishable on AUC but differ by $20.5M on profit because their profit-optimal thresholds approve slightly different fractions of the applicant pool. A study that selected on AUC alone would have chosen differently. Second, external macroeconomic and disaster variables contribute most of their value as main effects (HPI growth, real-income growth, hurricane exposure); the engineered interaction features add interpretability at the cost of negligible profit. Third, the largest single source of model performance is the raw application file. The SBA-only specification already reaches AUC 0.9809 and yields $1,087.9M, a starting point that constrains how much external data could realistically add.

For the decision problem studied here, the recommended policy is: use the Optuna-tuned LightGBM and approve applicants with predicted default probability at or below the profit-maximizing threshold.. Future work should test the framework on newer cohorts (a post-pandemic 2020–2024 sample would directly answer the single-crisis concern in Section 9.6), replace the stylized payoff matrix with loan-level revenue and recovery estimates, and run a formal fairness audit across state and NAICS sector before any deployment claim is made.

APPENDIX A: FULL MODEL RESULTS

Table 20 reports validation metrics for the 21 benchmark model configurations on the 44-feature cleaned set (554,292 training, 138,573 validation), followed by the separate Optuna-tuned LightGBM result. All thresholds were tuned to maximize net profit on the validation set. The profit function uses a 5% gain and 25% loss on DisbursementGross.

Table 20. Complete validation results for the 21 benchmark models plus the separate Optuna-tuned LightGBM follow-up.

Model	Profit	AUC	τ	Appr.	DR	ROI	Time
LGB Optuna (n418 nl194)	$1,096.4M	0.9827	0.1488	77.8%	1.09%	4.70%	12s
LGB n300 nl127	$1,078.6M	0.9814	0.1934	78.6%	1.39%	4.56%	12s
LGB n300 nl63	$1,072.3M	0.9803	0.1786	77.9%	1.35%	4.58%	8s
Bagging n100	$1,062.3M	0.9774	0.1736	76.9%	1.31%	4.63%	54s
XGB n300 d10	$1,058.1M	0.9793	0.1439	77.1%	1.29%	4.57%	6s
LGB n200 nl31	$1,055.1M	0.9772	0.1587	76.4%	1.32%	4.61%	4s
XGB n200 d8	$1,045.5M	0.9770	0.1637	76.9%	1.46%	4.52%	3s
HGB i300 d63	$1,044.4M	0.9776	0.1835	77.9%	1.61%	4.47%	27s
HGB i200 d31	$1,032.2M	0.9745	0.1667	76.5%	1.57%	4.48%	13s
AdaBoost n100	$1,020.0M	0.9712	0.4562	75.9%	1.71%	4.46%	132s
MLP 256×3	$978.5M	0.9595	0.0992	75.3%	2.24%	4.28%	91s
MLP 256×2	$972.3M	0.9573	0.1667	77.3%	2.68%	4.13%	97s
MLP 128×2	$970.9M	0.9580	0.1240	76.1%	2.42%	4.24%	70s
RF n100 d16 l25	$947.2M	0.9541	0.1667	73.2%	2.23%	4.10%	8s
RF n100 d16 l50	$936.0M	0.9521	0.1538	71.5%	2.10%	4.14%	8s
RF n200 d16 l50	$934.8M	0.9520	0.1538	71.5%	2.10%	4.13%	16s
Ridge C=0.1	$790.8M	0.8745	0.2182	72.4%	4.99%	3.34%	13s
Ridge C=10	$791.3M	0.8745	0.2232	73.0%	5.07%	3.33%	13s
Ridge C=1.0	$791.4M	0.8745	0.2232	72.9%	5.08%	3.34%	14s
LDA	$769.3M	0.8606	0.2083	71.6%	5.62%	3.32%	7s
QDA r=0.1	$697.4M	0.7944	0.0893	57.8%	6.01%	3.29%	5s
QDA r=0.2	$693.0M	0.7887	0.1042	59.0%	6.23%	3.21%	5s

Appr. = approval rate. DR = approved default rate. ROI = net profit / approved exposure. MLP 256×3 = 256-128-64-32 architecture. RF d = max_depth, l = min_samples_leaf.

APPENDIX B: MODEL PERFORMANCE CHARTS

Figures 19–22 provide visual comparisons across model families to complement the numerical results in Appendix A and the leaderboard in Section 6.1. All charts are generated from validation-set predictions on the 44-feature cleaned dataset.

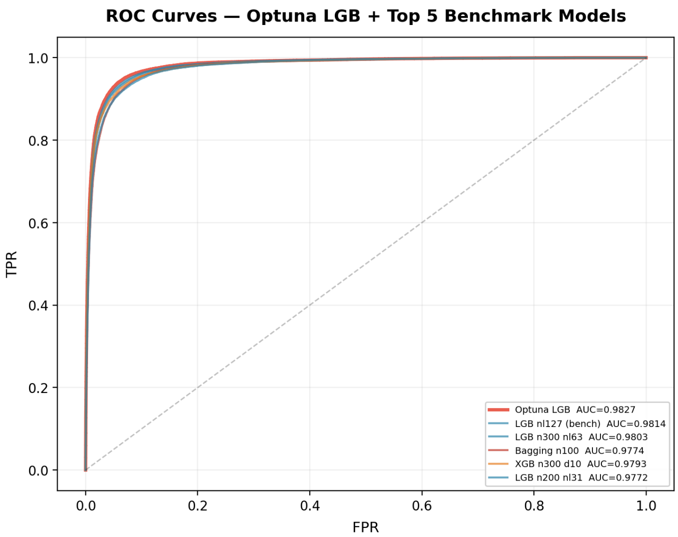

Figure 19. ROC curves for the Optuna-tuned LightGBM (highlighted) and the top five benchmark models by validation profit. LightGBM n300 nl127 (AUC 0.9814) and XGBoost n300 d10 (AUC 0.9793) are nearly indistinguishable in discriminative power, consistent with the finding that their profit gap arises from calibration differences near the decision threshold rather than ranking quality.

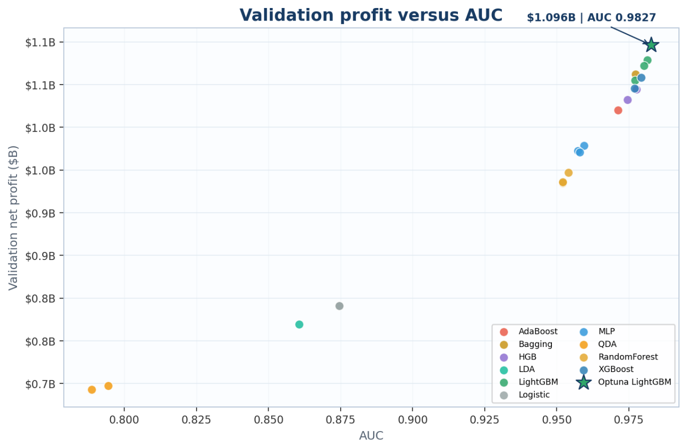

Figure 21. Profit versus AUC for the 21 benchmark model configurations. The Optuna-tuned LightGBM ($1,096.4M, AUC 0.9827) is shown as a highlighted star above the benchmark cluster. Tree-based ensembles occupy the upper-right region. The figure shows why model selection in lending should follow the profit objective rather than AUC alone.

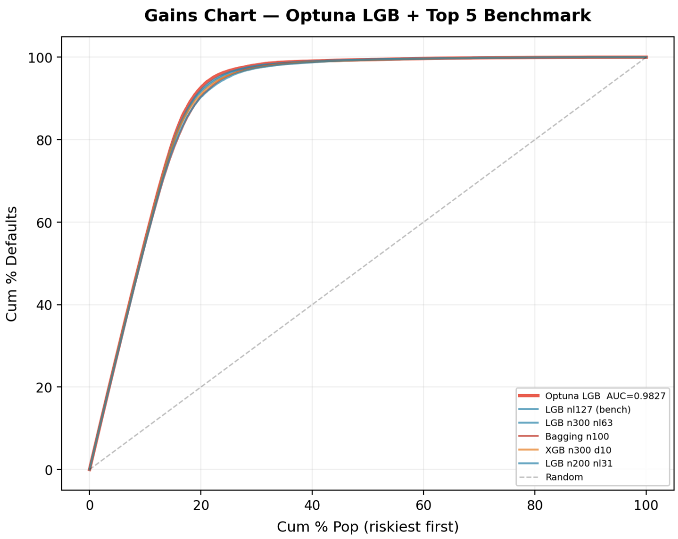

Figure 20. Gains chart for the Optuna-tuned LightGBM (highlighted) and the top five benchmark models. All models concentrate defaults in the riskiest tail; the top 20% of riskiest applicants contain roughly 70% of all defaults.

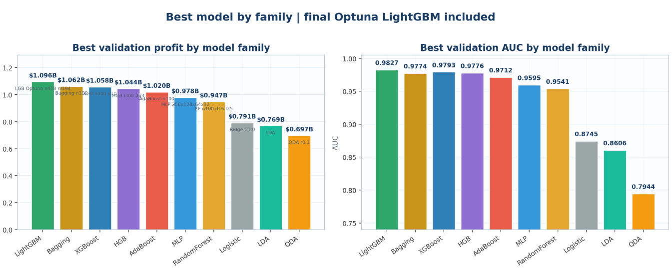

Figure 22. Best profit and AUC by model family. The Optuna-tuned LightGBM ($1,096.4M, highlighted) extends the LightGBM family beyond the benchmark entry. The gap between tree-based families and linear/neural families is visible in both panels. MLP bridges the gap on AUC (0.960) but not on profit.

Table 21. Complete feature glossary for the final 44-feature modeling set. Features are grouped by source: SBA-derived (16), external macro (15), and engineered interactions (13).

Feature	Type	Description
		SBA-derived baseline features (9 numerical + 7 categorical = 16)
Term	Numeric	Loan term in months. Range 0–300+; values ≥240 months proxy for real estate collateral.
log_GrAppv	Numeric	ln(1 + GrAppv), log-transformed gross approval amount to compress right-tail skew.
Portion	Numeric	SBA_Appv / GrAppv, the SBA guarantee share of the loan. Range (0, 1]; higher values shift default risk to SBA.
CreateJob	Numeric	Number of new jobs the borrower expects to create with loan proceeds.
RetainedJob	Numeric	Number of existing jobs the borrower expects to retain.
log_NoEmp	Numeric	ln(1 + NoEmp), log-transformed number of employees. Captures firm size.
jobs_per_dollar	Numeric	(CreateJob + RetainedJob) / GrAppv, employment-generation efficiency per dollar lent.
same_state_bank	Binary	1 if borrower's state = bank's state, 0 otherwise. Proxies local lending relationships.
RealEstate	Binary	1 if Term ≥ 240 months, 0 otherwise. Proxies real estate collateral backing.
State	Categorical	Borrower's state (2-letter abbreviation, 51 levels).
BankState	Categorical	Originating bank's state (2-letter abbreviation, 51 levels).
NewExist	Categorical	Business age: 1 = existing business, 2 = new business.
UrbanRural	Categorical	Location type: 0 = undefined, 1 = urban, 2 = rural.
NAICS_sector	Categorical	2-digit NAICS industry code (20 levels).
LowDoc_clean	Binary	1 if loan originated under the SBA LowDoc loan program (streamlined documentation), 0 otherwise.
RevLineCr_clean	Binary	1 if the loan includes a revolving line of credit facility, 0 otherwise.
		External macro indicators (15 features)
state_hpi_negative_flag	Binary	1 if the borrower's state 1-year HPI growth is negative at origination, 0 otherwise.
state_hpi_growth_1y	Numeric	Year-over-year percentage change in state-level FHFA House Price Index in the 12 months before approval.
state_hpi_drawdown_2y	Numeric	Maximum peak-to-trough decline in state HPI over the 24 months before approval. Captures housing stress severity.
high_inflation_flag_lag1	Binary	1 if year-over-year CPI change in the 12 months before approval exceeds the 90th percentile of the full-sample distribution.
cpi_yoy_change_lag1	Numeric	Year-over-year percentage change in the Consumer Price Index (CPI-U, all items) in the 12 months before approval.
oil_shock_flag_lag1	Binary	1 if year-over-year crude petroleum PPI change exceeds its 90th percentile.
oil_yoy_change_lag1	Numeric	Year-over-year percentage change in crude petroleum prices in the 12 months before approval.
real_income_decline_flag_lag1	Binary	1 if state-level real personal income growth over the prior year is negative.
real_income_growth_1y_lag1	Numeric	Real (inflation-adjusted) state personal income growth rate over the year before approval.
fema_hurricane_12m	Binary	1 if FEMA declared a hurricane disaster in the borrower's state within 365 days before approval.
fema_fire_12m	Binary	1 if FEMA declared a fire disaster in the borrower's state within 365 days before approval.
fema_flood_12m	Binary	1 if FEMA declared a flood disaster in the borrower's state within 365 days before approval.
fema_severe_storm_12m	Binary	1 if FEMA declared a severe storm disaster in the borrower's state within 365 days before approval.
state_bank_failures_12m	Numeric	Number of FDIC-insured bank failures in the borrower's state in the 12 months before approval.
industry_contraction_flag	Binary	1 if state-industry employment growth (BLS QCEW) was negative in the year before approval.
		Engineered interaction features (13 features)
infl_micro_biz	Binary	$\mathbb{1}[\texttt{high\_inflation\_flag} = 1] \times \mathbb{1}[\texttt{log\_NoEmp} \leq \ln(6)]$. Micro firms (0–5 employees) under high inflation.
oil_shock_supply	Binary	$\mathbb{1}[\texttt{oil\_shock\_flag} = 1] \times \mathbb{1}[\texttt{NAICS} \in \{23, 31, 32, 33, 48, 49\}]$. Fuel-exposed sectors during oil-price spikes.
income_decl_consumer	Binary	$\mathbb{1}[\texttt{real\_income\_decline\_flag} = 1] \times \mathbb{1}[\texttt{NAICS} \in \{44, 45, 71, 72\}]$. Consumer-facing sectors when real income is declining.
recent_bank_failure	Binary	$\mathbb{1}[\texttt{state\_bank\_failures\_12m} > 0]$. Proxies credit-cycle tightening in the borrower's state.
real_estate_industry_hpi	Binary	$\mathbb{1}[\texttt{NAICS} = 53] \times \mathbb{1}[\texttt{state\_hpi\_negative\_flag} = 1]$. Real estate sector (NAICS 53) during housing-price declines.
construction_hpi	Binary	$\mathbb{1}[\texttt{NAICS} = 23] \times \mathbb{1}[\texttt{state\_hpi\_negative\_flag} = 1]$. Construction sector during housing-price declines.
supply_chain_infl	Binary	$\mathbb{1}[\texttt{NAICS} \in \{23, 31, 32, 33, 42, 48, 49\}] \times \mathbb{1}[\texttt{high\_inflation\_flag} = 1]$. Supply-chain-exposed sectors under high inflation.
tourism_flag	Binary	$\mathbb{1}[\texttt{NAICS} \in \{44, 45, 71, 72\}]$. Tourism-dependent sectors (retail, hospitality, arts).
coastal_flag	Binary	$\mathbb{1}[\texttt{State} \in \text{Atlantic/Gulf}]$. 18 Atlantic and Gulf Coast states.
dsast_sensitive	Binary	$\mathbb{1}[\texttt{NAICS} \in \{23, 44, 45, 48, 49, 51, 71, 72\}]$. Disaster-sensitive sectors.
tourism_hurricane	Binary	tourism_flag × hurricane. Tourism-sector firms after hurricane exposure.
coastal_hurricane	Binary	coastal_flag × hurricane. Coastal-state firms after hurricane exposure.
dsast_hurricane	Binary	dsast_sensitive × hurricane. Disaster-sensitive-sector firms after hurricane exposure.

Features are listed in the order they appear in the model matrix. All binary flags take values $\{0, 1\}$. All external variables are constructed using only information available at or before the loan approval date.

APPENDIX C: FEATURE GLOSSARY

Table 21 provides definitions for every feature in the final 44-feature modeling set, organized by source. The table is designed to help readers interpret feature-importance rankings (Fig. 15), SHAP summary plots (Fig. 16), and the EDA interaction charts that reference these variables.

APPENDIX D: DATA SOURCES

The SBA 7(a) loan origination data used in this study is publicly available from the U.S. Small Business Administration data portal (data.sba.gov). The dataset spans 1987–2014 and contains 866,082 loans across 100 application-level columns including loan terms, amounts, borrower characteristics, lender geography, industry codes, and loan performance outcomes.

Seven external datasets were merged at origination using state, year, and date-based lookback windows as described in Section 3.3. All datasets are publicly accessible:

FHFA House Price Index. State-level all-transactions HPI (seasonally adjusted). Federal Housing Finance Agency. https://www.fhfa.gov/DataTools/Downloads/Pages/House-Price-Index.aspx

BLS Consumer Price Index. CPI for All Urban Consumers (CPI-U), national, all items, not seasonally adjusted. U.S. Bureau of Labor Statistics. https://www.bls.gov/cpi/data.htm

EIA Petroleum Prices. Crude petroleum PPI and spot prices. U.S. Energy Information Administration. https://www.eia.gov/petroleum/data.php

BEA Personal Income. State-level personal income, quarterly. U.S. Bureau of Economic Analysis. https://www.bea.gov/data/income-saving/personal-income-by-state

FDIC Bank Failures. Failed bank list with closure dates and locations. Federal Deposit Insurance Corporation. https://www.fdic.gov/resources/resolutions/bank-failures/failed-bank-list/

FEMA Disaster Declarations. OpenFEMA dataset of disaster declarations by type, state, and date. Federal Emergency Management Agency. https://www.fema.gov/openfema-data-page/disaster-declarations-summaries-v2

BLS QCEW. Quarterly Census of Employment and Wages, state-industry employment and wage data. U.S. Bureau of Labor Statistics. https://www.bls.gov/cew/downloadable-data-files.htm

All external variables were constructed using lookback windows ending strictly before the loan approval date, ensuring no post-decision information leakage. Flags for high inflation and oil shocks were defined at the 90th percentile of their respective distributions over the full sample period, as described in Section 3.3. The QCEW variables, except for the industry contraction flag which is retained as a model feature, were used as exploratory diagnostics and are noted as a direction for future work.

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